

QuickCell-12: A Multi-Stage Pruning Framework for Scalable SiDB Standard-Cell Synthesis

I. INTRODUCTION & MOTIVATION

While the flaws inherent in Moore's law are increasingly being recognized, the new paradigm of Field-coupled Nanocomputing (FCN) has come one step closer to replacing the old CMOS paradigm as its successor. The reason why the FCN paradigm constitutes a paradigm shift is that it entails a fundamentally different form of computing. In particular, the FCN paradigm is concerned with the interaction between atoms within electric fields rather than charge carrier flow. One of the most promising implementations of the FCN paradigm pertains to the application of Silicon Dangling Bonds (SiDBs). Indeed, it has been achieved some impressive breakthroughs in the fabrication of the hydrogen-passivated silicon surface.

While the conventional CMOS logic involves implementing the Boolean logic using a network of transistors, the SiDB involves making use of atomic sites that are similar to quantum dots that have different states of charges. As such, one is able to implement Boolean logic operations such as OR, AND, NAND, XOR, amongst others. In the conventional implementation method, Boolean logic operations need to be present as a physically implemented unit having pre-designed input and output structures. In the case of SiDB, it entails designing logic circuits from scratch using pre-designed input/output skeletons such that the final output atomic sites represent the intended Boolean logic operation.

However, defining such layout configurations is extremely difficult for a number of reasons:

1. Combinatorial explosion of candidate layout configurations:

Given $|C|$ admissible locations within the canvas and d canvas SiDBs to construct, the number of candidate layouts is of order $\binom{|C|}{d}$

2. Extreme sparsity of candidate layout implementations:

Indeed, only a small percentage of candidate layouts would satisfy the desired Boolean function. Consequently, much of the computing resources would end up being wasted on invalid candidates.

3. High overhead of physical verification:

To determine whether or not a certain layout design satisfies the required properties, it becomes necessary to test the function using all possible configurations of input patterns. This requires up to 2^n evaluations for each layout, where n refers to the number of input pins. Additionally, according to research findings, the testing procedure for a particular layout has proven to be an extremely time-intensive process, which can be highly exponential in relation to the total number of SiDBs employed in a given circuit layout design.

4. Difficulties with increasing number of inputs:

With an increase in the number of I/O pins, the complexity of the skeleton increases as well as the pattern of electrostatic interactions. Additionally, the cost of performing the physical simulation increases significantly.

However, owing to these limitations, the designing of circuit layouts for SiDB circuits is considerably behind the state of development reached by the layout design of traditional CMOS circuits, where technology has progressed beyond simple two-input gate designs into organized collections of complex multi-input multi-output gates. To overcome this problem, more than just a faster simulation tool will be required; rather, a proper selection procedure for invalid layouts before simulation is essential.

With the aim of dealing with the above challenge, the current paper proposes a superior design approach which elevates the concept of the pruning technique as used in QuickCell into a completely different paradigm. While the previous approach employed by QuickCell proved the concept of the use of pruning techniques to significantly reduce the size of the search space, the current research generalizes this idea through the application of three-stage twelve filter approach. As opposed to the previous research which utilized three special pruning techniques, the current paper utilizes the following hierarchical twelve filters such as:

- **Phase 1: Geometric and Structural Pruning,**
which eliminates layouts that are invalid due to minimum-distance violations, symmetry redundancy, wire interference, or the potential occurrence of positive charge states;
- **Phase 2: Analytical Physics-Based Pruning,**
which delete layouts that fail necessary conditions related to charge-count feasibility, input-pin integrity, electrostatic path connectivity, output-state attainability, directional output polarization, or analytical energetic consistency
- **Phase 3: Enumerative Physical Pruning,**
which applies explicit canvas-state enumeration to remove layouts that are physically infeasible under the required I/O constraints or energetically unstable with respect to competing incorrect I/O assignments.

Physical simulation must be performed with precision only if these considerations are kept in mind. The fundamental principle on which this whole approach is based is that there must not be any assault on the search space itself. Instead, pruning of the search space must start with the cheapest method right up to its most expensive one, starting with invalid layouts. Only a few layouts out of all of them must be allowed access to this expensive technique.

The following outcomes can be expected from applying the presented framework (3 phase/12 filters):

1. Significant decrease in the size of search space when compared to 3 filter solution
2. Decrease in the required time for the algorithm execution

3. Scalability to a more complicated functionality of SiDB standard cells, including multi-inputs cells
4. Number of precise physical simulations will be significantly decreased; consequently, extending the capabilities of automatic generation of SiDB standard cells.

Thus, the current model enhances the design of the QuickCell technology by considering the primary principle of this technique which is based on the use of pruning in order to develop a good logic design of the SiDB cell. Improvement is achieved by switching from three filters to twelve filters design.

II Methodology :

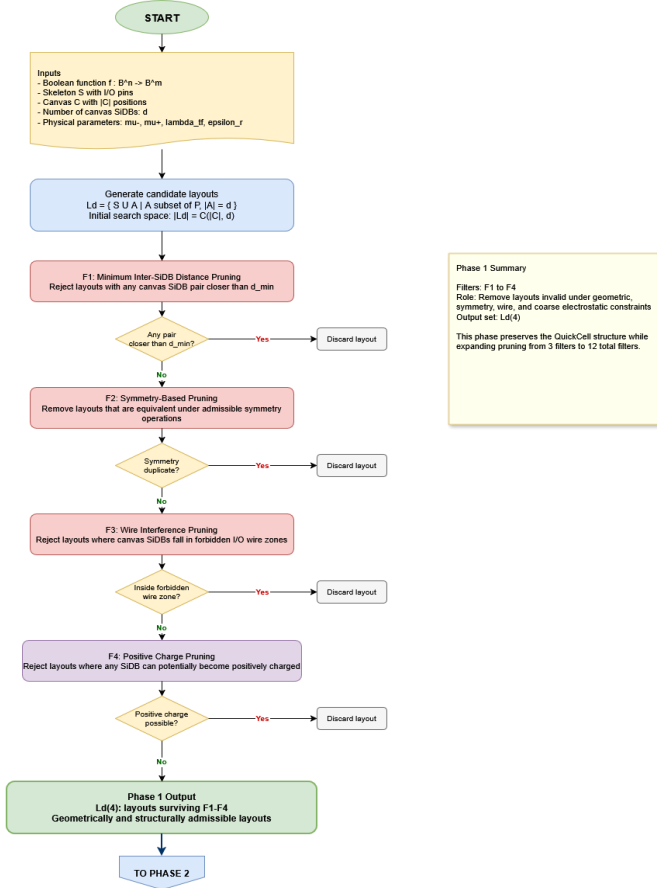
Here is the generalization of the QuickCell-12 approach presented in this section of this paper. It is clear from the previous analysis that the original approach to QuickCell had been able to show that, using good pruning techniques, it is possible to decrease the search space of SiDB standard cell design significantly. Nevertheless, the current research proposes to generalize this idea into an elaborate hierarchy. In contrast to the original concept of the QuickCell approach, the new technique includes twelve pruning filters in three phases, while the original concept included only three types of pruning techniques.

let the full search space be $L_d = \{ S \cup A \mid A \subseteq P, \mid A \mid = d \}$, where S is the fixed skeleton, P is the set of admissible canvas positions, and d is the number of canvas SiDBs. The twelve-filter framework defines the nested sequence $L_d \supseteq L_d^{(4)} \supseteq L_d^{(10)} \supseteq L_d^{(12)} \supseteq L^*$, where: $L_d^{(4)}$ is the output of Phase 1, $L_d^{(10)}$ is the output of Phase 2, $L_d^{(12)}$ is the output of Phase 3, and L^* is the final set of valid standard-cell implementations after exact physical verification.

The remainder of this section is organized as follows. First, in Phase 1: Geometric and Structural Pruning is presented, followed by Phase 2: Analytical Physics-Based Pruning, and finally Phase 3: Enumerative Physical.

II A. Phase 1: Geometric and Structural Pruning

Phase 1: Geometric and Structural Pruning (F1-F4)
QuickCell upgraded from 3 filters to 12 filters



Phase1_Prune

```

1: Input: Layout  $L$ , Skeleton  $S$ , Canvas  $C$ , Number of canvas SiDBs  $d$ , Physical parameters  $P = \{\mu_-, \mu_+, \lambda_{tf}, \epsilon_r\}$ 
2: Output: TRUE iff layout  $L$  must be discarded in Phase 1; FALSE otherwise
3: // F1: Minimum Inter-SiDB Distance Pruning
4: if F1_MinDistancePruning( $L, d, P$ ) = TRUE then
5:
6:     return TRUE
7: end if
8: // F2: Symmetry-Based Pruning
9: if F2_SymmetryPruning( $L, S, C$ ) = TRUE then
10:
11:     return TRUE
12: end if
13: // F3: Wire Interference Pruning
14: if F3_WireInterferencePruning( $L, S, P$ ) = TRUE then
15:
16:     return TRUE
17: end if
18: // F4: Positive Charge Pruning
19: if F4_PositiveChargePruning( $L, P$ ) = TRUE then
20:
21:     return TRUE
22: end if
23:
24: return FALSE

```

The first stage rejects layout configurations that fail on geometric, combinatorial, or crude electrostatic grounds. The filters at this stage are inexpensive and are used to reject poor layout designs that are easily identifiable without performing any electrostatic calculations.

The four filters in this phase are:

1. F1: Minimum Inter-SiDB Distance Pruning
2. F2: Symmetry-Based Pruning
3. F3: Wire Interference Pruning
4. F4: Positive Charge Pruning

Collectively, these filters transform the original search space L_d into $L_d^{(4)} = F_4 (F_3(F_2(F_1(L_d))))$

Each of these filters is briefly summarized below

II A.1- F1: Minimum Inter-SiDB Distance Pruning

F1_MinDistancePruning

```

1: Input: Layout  $L$ , Number of canvas SiDBs  $d$ , Physical
   parameters  $(\Theta)$ 
2: Output: TRUE iff any pair of canvas SiDBs violates
    $d_{\min}$ ; FALSE otherwise
3:  $A \leftarrow$  set of canvas SiDBs in  $L$ 
4:  $d_{\min} \leftarrow \text{ComputeMinimumAllowedDistance}(\Theta)$ 
5: for all pair  $(a_i, a_j)$  in  $A$  with  $i \neq j$  do
6:   if  $\text{EuclideanDistance}(a_i, a_j) < d_{\min}$  then
7:
8:     return TRUE
9:   end if
10: end for
11:
12: return FALSE

```

Physical parameters $\Theta = \{\mu-, \mu+, \lambda, \text{TF}, \epsilon, r\}$; q_e and ϵ_0 are physical constants.

1. Probabilistic and Combinatorial Interpretation

Filter F1 discards all those layouts for which the distance between at least one pair of the SiDBs on the canvas A (A = selected SiDBs on the canvas) is smaller than the smallest admissible distance, $d_{\min}$.

Although the condition may seem straightforward on the surface, it actually carries great physical and combinatorial meaning. In reality, the collection of all those layouts which meet the criteria of the first filter, namely the set $L_d^{(1)}$, can be formally defined using graph theory and probabilistic combinatorics.

Since the skeleton is fixed and assumed to be revalidated, and since the admissible canvas positions are defined so as not to collide with the skeleton, F1 is applied only to pairs of selected canvas SiDBs.

Let us define a conflict graph $G_{\text{conflict}} = (P, E_{\text{conflict}})$, where the vertex set P is the set of all $|C|$ admissible canvas positions, and an edge $(\vec{p}_i, \vec{p}_j) \in E_{\text{conflict}}$ exists if and only if the Euclidean distance between the two positions is less than the threshold: $\|\vec{p}_i - \vec{p}_j\|_2 < d_{\min}$

In this formulation, an edge represents a conflict, meaning two positions cannot simultaneously be occupied by SiDBs. A valid selection of d canvas positions, $A \subseteq P$, that satisfies the F1 criterion must therefore form an independent set of size d in the conflict graph G_{conflict} . An independent set is a subset of vertices where no two vertices are connected by an edge.

Thus, the set of layouts that survive F1 can be expressed as $L_d^{(1)} = \{S \cup A \mid A \in \mathcal{I}_d(G_{\text{conflict}})\}$ where $\mathcal{I}_d(G_{\text{conflict}})$ is the set of all independent sets of size d in the graph G_{conflict} .

The problem of counting $|L_d^{(1)}|$ is equivalent to counting the number of independent sets of a fixed size in a geometric graph, which is a known #P-complete problem. However, for a given layout A , verifying membership in this set simply requires checking all $\binom{d}{2}$ pairs, leading to an efficient $O(d^2)$ complexity per layout.

2. Derivation of $d_{\min}$ from First Principles

It must be emphasized that the $d_{\min}$ value does not constitute an empirical estimation of any kind, but rather reflects the principles of physics associated with charge instability within the SiDB system, which is the minimum distance between the particles where an unstable positive charge can form, irrespective of optimal experimental conditions.

The condition for an SiDB at site i to become positively charged is given by the population stability criterion $\mu^+ + q_e V_{\text{local},i} > 0$. In order to obtain a conservative and robust filter, we consider the worst-case scenario, i.e., we maximize $V_{\text{local},i}$. This maximum value occurs when all the SiDBs around site i are negatively charged. To obtain a hard bound, we consider the most important contribution from a single SiDB located at site j , with a charge $n_j = -1$, and a distance d from site i . The potential at site i is

then $V(d) = (-1) \frac{q_e}{4\pi\epsilon_0\epsilon_r} \frac{e^{-\frac{d}{\lambda_{TF}}}}{d} = \frac{e}{4\pi\epsilon_0\epsilon_r} \frac{e^{-\frac{d}{\lambda_{TF}}}}{d}$ and The critical distance $d_{\min}$ is found by setting the

positive charge activation energy to zero, $\mu^+ + q_e V(d_{\min}) = 0 \implies V(d_{\min}) = -\frac{\mu^+}{q_e} = \frac{|\mu^+|}{e}$

Substituting the expression for $V(d)$ gives the fundamental transcendental equation for

$d_{\min}$: $\frac{e^2}{4\pi\epsilon_0\epsilon_r} \frac{e^{-d_{\min}/\lambda_{TF}}}{d_{\min}} = |\mu^+|$, This equation links the geometric constraint ($d_{\min}$) directly to the core physical parameters of the system

3. Closed-Form Solution via the Lambert W Function

The equation for $d_{\min}$ can be solved analytically to yield a closed form expression. Let us define the electrostatic energy scale constant $A = \frac{e^2}{4\pi\epsilon_0\epsilon_r}$, The equation becomes, $A \frac{e^{-d_{\min}/\lambda_{TF}}}{d_{\min}} = |\mu^+|$

Then, we get: $d_{\min} e^{d_{\min}/\lambda_{TF}} = \frac{A}{|\mu^+|}$ and By making the substitution $u = d_{\min}/\lambda_{TF}$, such that $d_{\min} =$

$u\lambda_{TF}$, we obtain $(u\lambda_{TF})e^u = \frac{A}{|\mu^+|} \implies ue^u = \frac{A}{\lambda_{TF}|\mu^+|}$ therefore This equation is in the canonical form

$xe^x = z$, whose solution is given by the Lambert W function, $x = W(z)$. Therefore, $u = W\left(\frac{A}{\lambda_{TF}|\mu^+|}\right)$ and , substituting back for $d_{\min}$, we arrive at the exact analytical solution:

$d_{\min} = \lambda_{TF} \cdot W\left(\frac{e^2}{4\pi\epsilon_0\epsilon_r\lambda_{TF}|\mu^+|}\right)$ where W is the principal branch of the Lambert W function. This relation gives an expression that holds physical significance for the threshold distance and hence lifts the filtering criterion to be a consequence of the system's quantum electrostatic nature. Application of this threshold distance is far more accurate theoretically than that of the former heuristic based one.

4. Completeness, Soundness, and Heuristic Nature of the Filter

Completeness of a pruning filter means that the filter does not remove any truly valid layout. Let $\mathcal{L}_{\text{valid}}$ denote the set of layouts that are physically realizable and functionally correct under the adopted physical model. A filter F_i is complete if

$$\forall L \in \mathcal{L}_{\text{valid}}: F_i(L) = \text{pass, or equivalently, } F_i(L) = \text{reject} \Rightarrow L \notin \mathcal{L}_{\text{valid}}$$

Some filters in QuickCell-12 admit direct completeness proofs under the adopted model, while others are heuristic guards whose completeness depends on calibrated physical assumptions. In the following, we distinguish between formally complete filters, conditionally complete filters, and heuristic filters.

Filter	Explanation	Completeness Status
F1 Minimum distance	Complete if $d_{\min}$ is treated as a hard physical/design-rule constraint	Conditionally complete
F2 Symmetry pruning	Keeps one representative per admissible equivalence class	Complete
F3 Wire interference	Complete only if the forbidden zone is certified invalid	Conditionally complete
F4 Positive charge	Rejects layouts that can enter a forbidden positive-charge regime	Complete under no-positive-charge design rule
F5 Charge-count bound	Complete only if calibrated bounds contain all valid layouts	Heuristic / conditionally complete
F6 Input disturbance	Complete only if r_{in} is a certified safe bound	Heuristic / conditionally complete
F7 Path connectivity	Complete only under a finite-range/epsilon-reachability assumption	Heuristic / conditionally complete
F8 Output potential bound	Interval bound excludes target stability window	Complete
F9 Output pressure bound	Pressure interval cannot contain required sign	Complete
F10 Energy reference bound	Current implementation uses all-neutral reference, not a certified bound	Heuristic / conditionally complete
F11 Physical infeasibility	No metastable supporting canvas state exists	Complete in strict mode
F12 I/O instability	Incorrect output has lower energy than correct output	Complete in strict mode

Theorem 1 (Conditional Completeness of Minimum-Distance Pruning)

Let $d_{\min}$ be a certified hard minimum separation threshold for selected canvas SiDBs. Assume that any layout containing two selected canvas SiDBs $a_i, a_j \in A$ with $\|a_i - a_j\|_2 < d_{\min}$

is physically inadmissible under the adopted SiDB design rules, because such a pair can activate an invalid charge regime or violates the minimum allowed placement constraint. Then F1 is complete.

Proof

F1 rejects a layout $L = S \cup A$ only if there exist two selected canvas SiDBs $a_i, a_j \in A$ such that

$$\|a_i - a_j\|_2 < d_{\min}$$

By the hard-separation assumption, any layout containing such a pair is outside the admissible physical design space and therefore cannot belong to $\mathcal{L}_{\text{valid}}$. Hence $F_1(L) = \text{reject} \Rightarrow L \notin \mathcal{L}_{\text{valid}}$

Therefore, F1 cannot reject any valid layout under the certified minimum-distance rule

II A.2- F2: Symmetry-Based Pruning

F2_SymmetryPruning

```
1: Input: Layout  $L$ , Skeleton  $S$ , Canvas  $C$ , Booleanfunction  $f$ 
2: Output: TRUE iff  $L$  is not the canonical representative
   of its symmetry class; FALSE otherwise
3:  $G_{\text{adm}} \leftarrow$  admissible symmetry group of the instance  $(S, C, f)$ 
4:  $\text{enc}_A \leftarrow \text{EncodeCanvasSubset}(A, C)$ 
5:  $\text{enc}_{\min} \leftarrow \text{enc}_A$ 
6: for all  $g \in G_{\text{adm}}$  do
7:    $A_g \leftarrow \text{ApplySymmetry}(g, A)$ 
8:    $\text{enc}_g \leftarrow \text{EncodeCanvasSubset}(A_g, C)$ 
9:   if  $\text{enc}_g < \text{enc}_{\min}$  then
10:     $\text{enc}_{\min} \leftarrow \text{enc}_g$ 
11:   end if
12: end for
13: if  $\text{enc}_A \neq \text{enc}_{\min}$  then
14:
15:   return TRUE
16: end if
17:
18: return FALSE
```

1. Group-Theoretic Foundation

The second filter, F2, removes redundant layouts arising from admissible geometric symmetries of the standard-cell instance. Let $P \subset \mathbb{R}^2$ denote the set of admissible canvas positions, and let $A \subseteq P$ be a selected set of d canvas SiDBs. In contrast to considering the symmetry group of the canvas alone, the implementation uses the admissible symmetry group $G_{\text{adm}}(S, P, f)$ of the complete design instance. A transformation g belongs to G_{adm} only if it maps the canvas position set onto itself, preserves the

skeleton structure, maps input and output pins to compatible pins while preserving BDL orientation, preserves internal wire geometry, and leaves the Boolean function invariant under the induced input/output permutation.

Under this admissible group action, two canvas selections A_1 and A_2 are considered equivalent if there exists $g \in G_{adm}$ such that $A_2 = g(A_1)$. Each equivalence class forms an orbit, and only one representative from each orbit needs to be evaluated. This representative is chosen by canonicalization.

2. Canonical Labeling and Pruning

To implement the filter, we must define a canonical form, $\text{canonical}(A)$, for any layout A . The canonical form is a unique representative chosen from its orbit. A common and robust method is to use a lexicographical ordering. Each layout A can be represented as a sorted vector of the indices of its constituent canvas positions. The canonical form is then defined as the lexicographically smallest layout in its orbit: $\text{canonical}(A) = \min_{\text{lex}} \{g(A) \mid g \in G_{adm}\}$ where the comparison is performed on the vector representations of the layouts. The set of layouts that survive F2 is the set of all canonical layouts:

$$L_d^{(2)} = \{L = S \cup A \in L_d^{(1)} \mid A = \text{canonical}(A)\}$$

3. Completeness, Soundness, and Heuristic Nature of the Filter

Theorem 2 (Completeness of Symmetry Pruning)

Let G_{adm} denote the admissible symmetry group of a synthesis instance (S, C, f)

For a candidate layout (A) , define its orbit $\mathcal{O}(A) = \{g(A) \mid g \in G_{adm}\}$

Let the canonical representative of the orbit be $A_{can} = \min \mathcal{O}(A)$

with respect to the encoding order used by `Encode Canvas Subset(·)`

Then the symmetry pruning filter (F_2) is complete; namely, $F_2(A) = \text{reject}$, implies that another surviving layout exists that is physically and functionally equivalent to (A)

Proof

By definition, every transform $g \in G_{adm}$ satisfies the following criteria:

satisfies the following properties:

1. The transformed layout remains inside the canvas
2. Input pins are mapped onto input pins
3. Output pins are mapped onto output pins
4. The wire structure is invariant
5. The Boolean function is invariant under the corresponding input/output permutations

Therefore every admissible transformation preserves the synthesis problem

Let $A' = g(A)$, for some $g \in G_{adm}$

Since Euclidean distances are preserved under reflections and rotations, $d_{ij}(A)$, $d_{ij}(A')$

Hence all screened Coulomb interaction terms satisfy $\phi_{ij}(A)$, $\phi_{ij}(A')$

Consequently, $E(A, c)$, $E(A', c')$

for every corresponding charge configuration

Therefore the two layouts possess identical electrostatic behavior, identical stability properties, and identical logical functionality.

The orbit $\mathcal{O}(A)$, thus consists entirely of equivalent layouts

F2 retains exactly one element of this orbit: $A_{can} = \min \mathcal{O}(A)$

If a layout (A) is rejected, then $A \neq A_{can}$

Hence (A_{can}) remains in the search space and represents the same physical solution class as (A)

Therefore no unique valid solution is removed

Secondary result

Let $\mathcal{V}$ be the set of valid layouts Then $\forall A \in \mathcal{V}, F_2(A) = \text{reject}$

implies the existence of $A_{can} \in \mathcal{V}$, such that $A_{can} \sim A$,therefore symmetry pruning preserves completeness

II A.3- F3: Wire Interference Pruning

F3_WireInterferencePruning

```

1: Input: Layout  $L$ , Skeleton  $S$ , Physical parameters  $\Theta$ 
2: Output: TRUE iff any selected canvas SiDB lies
   inside the protected internal wire zone ; FALSE otherwise
3:  $A \leftarrow$  set of canvas SiDBs in  $L$ 
4:  $\Gamma_{int} \leftarrow$  protected internal wire geometry extracted from  $S$ 
5:  $r_{int} \leftarrow \text{ComputeWireInterferenceRadius}(\Theta)$ 
6: for all  $a \in A$  do
7:   if  $\text{DistanceToWireGeometry}(a, \Gamma_{int}) < r_{int}$  then
8:
9:     return TRUE
10:  end if
11: end for
12:
13: return FALSE

```

$\Theta = \{\mu-, \mu+, \lambda TF, \epsilon r\}$; r_{int} is implemented as $\text{wire_forbidden_radius}$, or as $\max(d_{min}, 0.5\lambda TF)$ if not explicitly specified.

1. Physical Foundation: Kink State Induction in BDL Wires

The third filter removes layouts whose canvas SiDBs are inserted into electrostatically protected neighborhoods around internal wire SiDBs that are explicitly stored in the skeleton.

Such placements can destabilize the charge encoding of the wires and induce unwanted kinks or overpopulation effects. F3 therefore removes any layout in which a canvas SiDB lies within a forbidden interference zone around the protected internal wire geometry, resulting in the set $L_d^{(3)}$

Or, we can say The third filter F3, is designed to preserve the signal integrity of the input/output (I/O) wires. These wires are typically implemented as chains of Binary-Dot Logic (BDL) pairs. A signal '1' or '0' propagates as a specific charge arrangement that travels down the chain. The stability of this propagation depends on a delicate balance of local electrostatic potentials.

A kink state is a localized bit-flip within the wire, where a BDL pair adopts a charge state opposite to the intended signal. For example, in a wire propagating a '1' signal (e.g., a sequence of (0,-1) pairs), a kink would be a pair in the (-1,0) state. This can disrupt or entirely block signal flow.

A canvas SiDB placed near an internal BDL wire pair acts as a significant local electrostatic perturber. The potential from this single canvas SiDB can be strong enough to invert the potential difference across a BDL pair, thereby inducing a kink. Let us consider an internal BDL wire pair p_k with dots at positions $\vec{r}_{k,a}$ and $\vec{r}_{k,b}$. The bit state is determined by the signs of the local potential difference, or electrostatic pressure: $\Delta V_k = V_{\text{local}}(\vec{r}_{k,a}) - V_{\text{local}}(\vec{r}_{k,b})$. The wire is designed such that without any external disturbance, the potential from the rest of the circuit, $\Delta V_{\text{circuit}}$, establishes the correct sign. A canvas SiDB at position $\vec{r}_c$ with charge $n_c = -1$ introduces an additional differential potential, $\Delta V_{c,k}$:

$\Delta V_{c,k}(\vec{r}_c) = V_{k,a;c}(\vec{r}_{k,a}, \vec{r}_c, -1) - V_{k,b;c}(\vec{r}_{k,b}, \vec{r}_c, -1)$ A kink is induced if this disturbance is sufficient to flip the sign of the total pressure:

$\text{sgn}(\Delta V_{\text{circuit}}) \neq \text{sgn}(\Delta V_{\text{circuit}} + \Delta V_{c,k})$ This occurs when $|\Delta V_{c,k}| > |\Delta V_{\text{circuit}}|$

The value $|\Delta V_{\text{circuit}} \cdot q_e|$ can be interpreted as the energy barrier to kink formation, ΔE_{kink}

2. Formal Definition of the Forbidden Interference Zone

F3 filter operates by defining a forbidden interference zone, $\mathcal{F}$, a region in $\mathbb{R}^2$ where the placement of a canvas SiDB is disallowed. This zone is the set of all points $\vec{r}_c$ where a single negative SiDB can induce a kink in any internal wire pair.

For each internal wire pair $p_k \in W_{\text{internal}}$, we define a forbidden region $\mathcal{F}_k$:

$\mathcal{F}_k = \left\{ \vec{r}_c \in \mathbb{R}^2 \mid |\Delta V_{c,k}(\vec{r}_c)| > \frac{\Delta E_{\text{kink},k}}{q_e} \right\}$ where $\Delta E_{\text{kink},k}$ is specific energy barrier for pair p_k . total

forbidden zone for the entire layout is the union of the zones for all internal wire pairs:

$\mathcal{F} = \bigcup_{p_k \in W_{\text{internal}}} \mathcal{F}_k$ The boundary of each $\mathcal{F}_k$ is a level set (iso-potential contour) of the differential potential function $|\Delta V_{c,k}(\vec{r}_c)|$. Due to the nature of Yukawa potential, this boundary is not a simple geometric shape. However, for implementation, it can be tightly approximated by a simpler geometric primitive, such as an ellipse or a "stadium" shape, aligned along the wire axis. The parameters of this shape are calibrated from detailed physical simulations and are fixed once to conservatively contain the true forbidden region.

For example, approximating $\mathcal{F}$ with a union of balls of radius $r_{\text{forbidden}}$ is a common simplification:

$$\mathcal{F}_{\text{approx}} = \bigcup_{\vec{w} \in W_{\text{internal}}} B(\vec{w}, r_{\text{forbidden}}) \cup \bigcup_i ((wi, wi + 1) \cup Tube([wi, wi + 1], r_{\text{forbidden}}))$$

where W_{internal} now represents the set of all SiDB positions within the internal wires

3. Pruning Condition and Set-Theoretic Formulation

F3 filter removes any layout where a canvas SiDB position falls within this pre-defined forbidden zone. Let $A = \{\vec{a}_1, \dots, \vec{a}_d\}$ be the set of canvas SiDB positions for a given layout.

Discard layout $S \cup A$ if $A \cap \mathcal{F} \neq \emptyset$. This means the layout is discarded if there exist at least one canvas SiDB $\vec{a}_i \in A$ such that $\vec{a}_i$ is inside the forbidden zone $\mathcal{F}$.

Set of layouts that survive F3, denoted $L_d^{(3)}$, can be formally expressed as:

$L_d^{(3)} = \{L = S \cup A \in L_d^{(2)} \mid A \cap \mathcal{F} = \emptyset\}$ This is equivalent to stating that the set of selected canvas positions A must be a subset of the "allowed" region, $P \setminus \mathcal{F}$.

The computational complexity of this filter is efficient. For each of the d canvas SiDBs, it performs a point-in-polygon or distance check against the geometric primitives defining $\mathcal{F}$. If $\mathcal{F}$ is approximated by a union of $|W_{\text{internal}}|$ balls, the complexity is $O(d \cdot |W_{\text{internal}}|)$, making it a very fast structural check that effectively preserves the physical integrity of the I/O signal paths.

4. Completeness, Soundness, and Heuristic Nature of the Filter

Theorem 3 (Conditional Completeness of Wire-Interference Pruning)

Let $\mathcal{F}_{\text{cert}}$ be a certified forbidden wire-interference region such that any layout with at least one selected canvas SiDB inside $\mathcal{F}_{\text{cert}}$ violates the required I/O wire integrity.

Formally, assume $A \cap \mathcal{F}_{\text{cert}} \neq \emptyset \implies S \cup A \notin \mathcal{L}_{\text{valid}}$. If F3 reject exactly those layouts satisfying $A \cap \mathcal{F}_{\text{cert}} \neq \emptyset$, then F3 is complete.

Proof

F3 rejects a layout only when there exists a selected canvas SiDB $a \in A$ such that $a \in \mathcal{F}_{\text{cert}}$. By the definition of the certified forbidden region, this implies that the layout violates the I/O wire-integrity requirement and therefore cannot be a valid layout. Hence, $F_3(L) = \text{reject} \implies L \notin \mathcal{L}_{\text{valid}}$.

Thus F3 does not remove any valid layout under the certified forbidden-zone assumption. In case the forbidden region implemented is an over approximation of the physical interference region, F3 should be considered as a heuristic approach and not as a complete filter since completeness can be guaranteed only when each of the points within the forbidden region is known to be invalid.

II A.4- F4: Positive Charge Pruning

F4_PositiveChargePruning

```
1: Input: Layout  $L$ , Physical parameters  $\Theta = \{\mu_-, \mu_+, \lambda_{\text{tf}}, \epsilon_r\}$ 
2: Output: TRUE iff any SiDB can potentially become positively charged; FALSE otherwise
3: for all SiDB site  $i$  in  $L$  do
4:    $V_{\text{local\_wc}}(i) \leftarrow 0$ 
5:   for all SiDB site  $j$  in  $L$  with  $j \neq i$  do
6:      $V_{\text{local\_wc}}(i) \leftarrow V_{\text{local\_wc}}(i) - \phi_{ij}$ 
   end for
7:   if  $\mu_+ + q_e \cdot V_{\text{local\_wc}}(i) > 0$  then
8:     return TRUE
9:   end if
10: end for
11:
12: return FALSE
```

Fourth filter, inherited from the original QuickCell framework, eliminates layouts where one or more SiDBs have the potential to be positively charged. The worst-case local electrostatic potential is calculated by setting all the SiDBs to the negative state. If this is sufficient to satisfy the transition condition for the positive charge state, the layout is eliminated.

The output of this first phase is therefore $L_d^{(4)}$, which contains layouts that are geometrically admissible, symmetry-reduced, wire-safe, and compatible with the intended neutral/negative SiDB charge-state regime.

1. Physical Principle and Justification

Fourth filter, F4, enforces a fundamental physical constraint of current n-doped SiDB logic systems: the absence of positively charged SiDBs ($n_j = +1$) in any stable configuration. While a theoretical possibility, positive charges are energetically unfavorable in n-doped silicon and are not utilized in existing logic schemes. Their potential presence is therefore treated as a failure mode.

F4 filter is designed to be a computationally inexpensive yet powerful method to discard any layout that possesses even the *potential* to harbor a positive charge. It does not need to determine the ground state; it only needs to prove that a path to a positive charge state exists.

An SiDB at site i can transition to a positive state if its local potential $V_{\text{local},i}$ becomes sufficiently high (less negative) to satisfy the population stability condition: $\mu^+ + q_e V_{\text{local},i} > 0$ where μ^+ is the electrochemical potential for the $0 \rightarrow +1$ transition.

To create a robust pruning conditions, we check for this possibility under the most extreme, "worst-case" scenario for positive charge formation. The local potential at site i is a linear superposition of

contributions from all other SiDBs $j \neq i$: $V_{\text{local},i} = \sum_{j \neq i} n_j V_{ij}^{\text{unit}}$ where $V_{ij}^{\text{unit}} = \frac{q_e}{4\pi\epsilon_0\epsilon_r} \frac{e^{-d_{ij}/\lambda_{TF}}}{d_{ij}}$

Since V_{ij}^{unit} is negative, the total potential $V_{\text{local},i}$ is maximized (becomes least negative) when the charges n_j of the surrounding SiDBs are as negative as possible. The absolute maximum local potential at site i , $V_{\text{local},i}^{\text{max}}$, is therefore achieved when we assume all other SiDBs $j \neq i$ are in the $n_j = -1$ state.

2. Formal Mathematical Formulation

Let a layout be given by $L = S \cup A$, For any SiDB site $\vec{r}_i \in L$, we define maximum local potential as:

$V_{\text{local},i}^{\text{max}} = \sum_{j \in L, j \neq i} (-1) \cdot \frac{q_e}{4\pi\epsilon_0\epsilon_r} \frac{e^{-|\vec{r}_i - \vec{r}_j|/\lambda_{TF}}}{|\vec{r}_i - \vec{r}_j|}$ This value can be calculated analytically for any given layout without enumerating charge states

This filter then checks if this maximum potential is sufficient to cross the positive charge threshold for any SiDB in the layout.

Pruning Condition:

A layout L is discarded if there exists at least one site $\vec{r}_i \in L$ for which the worst-case potential could activate a positive charge. Discard L if $\exists i \in L$ such that $\mu^+ + q_e V_{\text{local},i}^{\text{max}} > 0$. If this condition is satisfied, then it means layout is invalid since there is a physical path to an invalid state. If this condition is not satisfied for all SiDB, it means it is a proof that no positive charge can be formed under any possible charge configuration. Because even the best potential is not enough

The set of layouts surviving this filter, and thus completing Phase 1, is formally defined as:

$$L_d^{(4)} = \left\{ L \in L_d^{(3)} \mid \forall i \in L: \mu^+ + q_e \left(\sum_{j \in L, j \neq i} \frac{-q_e}{4\pi\epsilon_0\epsilon_r} \frac{e^{-d_{ij}/\lambda_{TF}}}{d_{ij}} \right) \leq 0 \right\}$$

3. Algorithmic Complexity and Role in the Pipeline

For a layout with $N = |L|$ total SiDBs, calculating $V_{\text{local},i}^{\max}$ for a single site takes $O(N)$ time. Since this must be done for all N sites, the total complexity of the F4 check for one layout is $O(N^2)$ and as $|L| = |S| + d$, this can be written as $O((|S| + d)^2)$.

Although this filter requires providing physical parameter values, its computational cost is polynomial and much lower than the exponential cost of the enumerative filters in Phase 3. By placing this filter at the end of Phase 1, we can guarantee that it operates on a set of layouts which have already been checked for geometric and structural integrity. This filter acts as a vital link between the geometric filters and the more complex analytical filters of Phase 2. Its position here is optimal because it is more costly than the preceding geometric filters but much less costly than the following enumerative filters.

4. Completeness, Soundness, and Heuristic Nature of the Filter

Theorem 4 (Completeness of Positive-Charge Pruning under the No-Positive-Charge Rule)

Assume that a valid SiDB logic layout must not admit a physically accessible positively charged SiDB state. Let V_i^{wc} denote the worst-case local potential at site i obtained by assigning all other SiDBs the negative charge state. If F4 rejects a layout L , then L is not valid.

Proof

F4 rejects L if there exists a site i such that $\mu^+ + q_e V_i^{wc} > 0$.

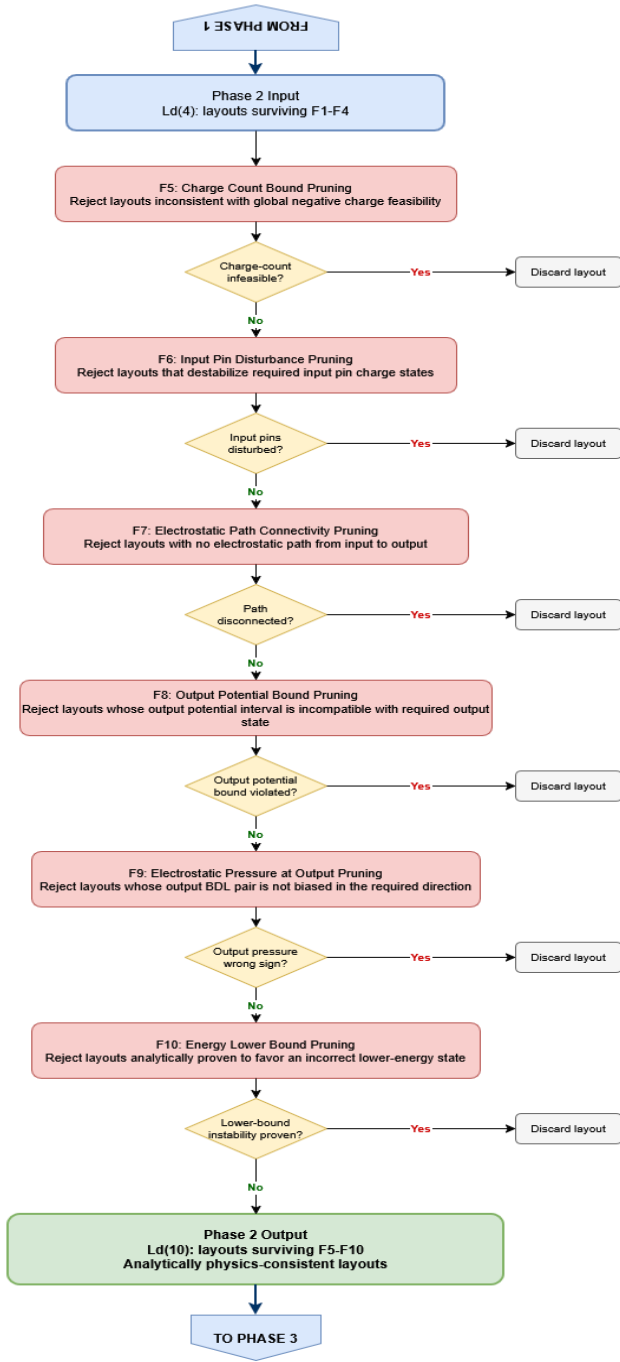
This is exactly the population-stability condition indicating that a positive charge state can become physically accessible at site i under the worst-case charge assignment. Since the design model excludes layouts that can admit positive charge states, such a layout cannot belong to $\mathcal{L}_{\text{valid}}$. Therefore,

$$F_4(L) = \text{reject} \implies L \notin \mathcal{L}_{\text{valid}}$$

Therefore F4 is complete with respect to the adopted no-positive-charge validity criterion.

II B. Phase 2: Analytical Physics-Based Pruning

Phase 2: Analytical Physics-Based Pruning (F5-F10) QuickCell upgraded from 3 filters to 12 filters



Phase 2 Summary

Filters: F5 to F10

Role: Remove layouts with charge-count, input-integrity, connectivity, output-feasibility, directional-bias, or analytical-energy issues

Output set: Ld(10)

This phase extends QuickCell using analytical electrostatic pruning before expensive enumeration.

```

Phase2_Prune
1: Input: Layout  $L$ , Boolean function  $f$ , Skeleton  $S$ ,
   Canvas  $C$ , Number of canvas SiDBs  $d$ , Physical
   parameters  $P$ 
2: Output: TRUE iff layout  $L$  must be discarded in Phase
   2; FALSE otherwise
3: // F5: Charge Count Bound Pruning
4: if F5_ChargeCountBoundPruning( $L, f, d, P$ ) =
   TRUE then
5:
6:   return TRUE
7: end if
8: // F6: Input Pin Disturbance Pruning
9: if F6_InputPinDisturbancePruning( $L, f, S, P$ ) =
   TRUE then
10:
11:   return TRUE
12: end if
13: // F7: Electrostatic Path Connectivity Pruning
14: if F7_ElectrostaticPathConnectivityPruning( $L, S, P$ ) =
   TRUE then
15:
16:   return TRUE
17: end if
18: // F8: Output Potential Bound Pruning
19: if F8_OutputPotentialBoundPruning( $L, f, S, P$ ) =
   TRUE then
20:
21:   return TRUE
22: end if
23: // F9: Electrostatic Pressure at Output Pruning
24: if F9_ElectrostaticPressurePruning( $L, f, S, P$ ) =
   TRUE then
25:
26:   return TRUE
27: end if
28: // F10: Energy Lower Bound Pruning
29: if F10_EnergyLowerBoundPruning( $L, f, S, P$ ) =
   TRUE then
30:
31:   return TRUE
32: end if
33:
   . . .

```

The filters used in Phase II are more complex compared to those of Phase I, with the reasoning being purely analytical from an electrostatic and energetic perspective. The filters used in Phase II are very different from the filter used in Phase I since the latter represents a claim regarding the viability of the circuit physically. However, like in Phase I, the filters in Phase II cannot validate every state of the canvas

All filters in Phase II are conservative necessary-condition filters. Surviving a filter does not imply functional correctness. Failure implies strong evidence of infeasibility

The six filters in this phase are: F5: Charge Count Bound Pruning ,F6: Input Pin Disturbance Pruning, F7: Electrostatic Path Connectivity Pruning , F8: Output Potential Bound Pruning ,F9: Electrostatic Pressure at Output Pruning , F10: Energy Lower Bound Pruning

Collectively, these filters transform the Phase-1 output into: $L_d^{(10)} = F_{10}(F_9(F_8(F_7(F_6(F_5(L_d^{(4)}))))))$

The six filters are summarized as follows:

II B.1- F5: Charge Count Bound Pruning

F5_ChargeCountBoundPruning

```

1: Input: Layout  $L$ , Boolean function  $f$ , Number of
   canvas SiDBs  $d$ , Physical parameters  $\Theta$ 
2: Output: TRUE iff required negative charge count is
   infeasible; FALSE otherwise
3: for all input pattern  $x \in \mathbb{B}^n$  do
4:    $N_{\text{req}} \leftarrow \text{RequiredSkeletonNegativeChargeCount}(x, f(x))$ 
5:    $I_{\text{logic}} \leftarrow [N_{\text{req}}, N_{\text{req}} + d]$ 
6:    $I_{\text{phys}} \leftarrow [\text{Nadm,min}, \text{Nadm,max}]$ 
7:   if  $I_{\text{logic}} \cap I_{\text{phys}} = \emptyset$  then
8:
9:     return TRUE
10:  end if
11: end for
12:
13: return FALSE

```

F5 verify that the total numbers of negative charges demanded by the imposed I/O encoding is consistent with the number of negative charges that are supported by the skeleton/canvas combination. If the logic-induced charge-count interval does not overlap with the heuristic admissible charge interval, the layout is considered infeasible by this global occupation-consistency criterion and is discarded

F5 is not intended to prove physical feasibility. It serves as a conservative occupation-consistency filter whose purpose is to eliminate obviously incompatible layouts before more rigorous analysis

This is how F5 filters out layouts based on a global occupation consistency check

1. Physical Motivation: Global Charge-Occupation Consistency

Motivation behind filter F5 stems from the observation that it is impossible to accommodate an unlimited number of negative charges in the configuration of an SiDB. This is because the achievable number of negatively charged SiDBs will depend on the electrostatic interactions between sites as well as the total number of chemical potential parameters. It is therefore clear that any layout which cannot have its number of negative charges adjusted to suit an imposed I/O encoding is unfeasible

In the present implementation, this physical constraint is modeled as a conservative configurable charge-count interval rather than by explicitly solving a grand-potential minimization problem. Specifically, for a layout L , the physically admissible interval is approximated as $N_{\text{adm}}(L, \Theta) = [[\beta \mid L \mid], [\alpha \mid L \mid]]$, where $\beta = \text{charge_min_fraction}$ and $\alpha = \text{charge_max_fraction}$ are user-defined or calibrated occupation-bound parameters. filter then checks whether this admissible interval overlaps with the logically required charge-count interval induced by each input/output assignment.

2. Formal Definition of Charge Count Intervals

a) Logically Required Charge Count Interval, $\mathcal{N}_{\text{req}}$

For a given input vector $\vec{x}$, the charge configuration of the I/O pin in the skeleton S is fixed. The number of negative charges on the skeleton, $N_{S,\text{req}}^-$, is determined by counting the number of BDL pairs that must hold a negative charge to represent the input $\vec{x}$ and the correct output $f(\vec{x})$

The d canvas SiDBs can each be either neutral ($n_j = 0$) or negative ($n_j = -1$), Therefore the number of negative charges on the canvas, N_A^- , can range from 0 to d

The total number of negative charges in the entire layout, N_{total}^- , must therefore lie within the following interval: $\mathcal{N}_{\text{req}}(x) = [N_{S,\text{req}}^-(x), N_{S,\text{req}}^-(x) + d]$

b) Physically Admissible Charge Count Interval, $\mathcal{N}_{\text{adm}}$

This interval, $[N_{\text{adm},\text{min}}^-, N_{\text{adm},\text{max}}^-]$, represents the minimum and maximum number of negative SiDBs the system can stably accommodate. Deriving these bounds with absolute rigor is complex, but for a pruning filter, we can establish effective, physically-motivated heuristic bounds.

- **Upper Bound ($N_{\text{adm},\text{max}}^-$):** As more SiDBs become negatively charged, the total Coulombic repulsion energy increases quadratically with N^- . At some point, the energy cost of adding another electron (due to repulsion) will exceed the energy gain from capturing an electron from the reservoir (related to μ^-). This sets an upper limit on N^- . A simple heuristic can be based on the average SiDB density, $\rho = N/\text{Area}$. Denser layouts support a lower fraction of negative charges. For a given layout family, this bound can be estimated empirically, e.g.,
$$N_{\text{adm},\text{max}}^- = \alpha \cdot |L|$$
, where $\alpha \in [0.5, 0.8]$ is a density-dependent factor.
- **Lower Bound ($N_{\text{adm},\text{min}}^-$):** The n-doped substrate creates a natural bias for SiDBs to become negatively charged. A system with too few negative charges is energetically unfavorable compared to one where more SiDBs have captured electrons from the reservoir. This sets a lower bound. For example, it might be physically implausible for a large, sparse layout to have fewer than $\beta \cdot |L|$ negative charges, where $\beta \in [0.2, 0.4]$.

For each specific gate topology (e. g., 3-input AND gate with a Y-shaped skeleton), these bounds $[N_{\text{adm},\text{min}}^-, N_{\text{adm},\text{max}}^-]$ can be pre-calibrated by analyzing the charge counts of known valid layouts.

3. Pruning Condition and Set-Theoretic Formulation

The F5 filter discards any layout where the range of negative charges required by the logic is completely disjoint from the range of negative charges the system can physically support.

For a given input vector $\vec{x}$: Discard layout L if $\mathcal{N}_{\text{req}}(\vec{x}) \cap \mathcal{N}_{\text{adm}}(L, \theta) = \emptyset$ This is equivalent to checking the boundary conditions, Discard if $(N_{S,\text{req}}^-(x) + d < N_{\text{adm},\text{min}}^-) \vee (N_{S,\text{req}}^-(x) > N_{\text{adm},\text{max}}^-)$

Since this check must hold for all input vectors, the layout survives only if the condition is false for all $\vec{x} \in \mathbb{B}^n$

The set of layouts that survive F5 is defined as $L_d^{(5)} = \{L \in L_d^{(4)} \mid \forall \vec{x} \in \mathbb{B}^n: \mathcal{N}_{\text{req}}(\vec{x}) \cap \mathcal{N}_{\text{adm}}(L) \neq \emptyset\}$

This filter is extremely fast, implementation runs in $O((2^n)|s|)$, which reduces to $O(2^n)$ for a fixed skeleton or when the required skeleton charge counts are precomputed.

It provides a powerful global consistency check, eliminating layouts that are fundamentally incompatible with the system's overall charge capacity before any detailed potential calculations are performed

4. Completeness, Soundness, and Heuristic Nature of the Filter

Theorem 5(Conditional Completeness of Charge-Count Bound Pruning)

Let $\mathcal{N}_{\text{adm}}(L, \Theta)$ be an admissible negative-charge-count interval that is certified to contain the total negative-charge count of every valid configuration of layout L. That is, assume

$L \in \mathcal{L}_{\text{valid}} \implies \forall x \in \mathcal{B}^n: \mathcal{N}_{\text{req}}(x) \cap \mathcal{N}_{\text{adm}}(L, \Theta) \neq \emptyset$, then F5 is complete

Proof

F5 rejects L only if there exists an input x such that , $\mathcal{N}_{\text{req}}(x) \cap \mathcal{N}_{\text{adm}}(L, \Theta) = \emptyset$

If $\mathcal{N}_{\text{adm}}$ is certified to contain all physically admissible negative-charge counts of valid layouts, then no valid layout can violate this intersection condition. Therefore any rejected layout is not valid. Hence F5 is complete under the certified charge-count interval assumption.

In the present implementation, $\mathcal{N}_{\text{adm}}$ is modeled by configurable heuristic bounds $[[\beta|L|], [\alpha|L|]]$. Therefore F5 is not claimed to be unconditionally complete unless these bounds are calibrated and certified for the considered layout family

II B.2-F6: Input Pin Disturbance Pruning

F6_InputPinDisturbancePruning

Input: Layout $L = S \cup A$, Boolean function f , Skeleton S , physical parameters Θ ,
input disturbance threshold τ_{in}
Output: TRUE iff canvas SiDBs excessively disturb an input pin; FALSE
otherwise

```

1: for all input pattern  $x \in B^n$  do       $y \leftarrow f(x)$ 
2:   for all input pin site  $i \in \text{InputPins}(S)$  do
3:      $Dwc(i) \leftarrow -\sum_{a \in A} \phi(i, a)$ 
4:     if enforce_input_population = FALSE then
5:       if  $|Dwc(i)| > \tau_{in}$  then
6:         return TRUE
7:       end if
8:     else
9:        $Vbase(i) \leftarrow \text{SkeletonIOPotential}(i, x, y, S, \Theta)$ 
10:       $Vdisturbed(i) \leftarrow Vbase(i) + Dwc(i)$ 
11:       $nreq(i) \leftarrow \text{RequiredInputCharge}(i, x)$ 
12:      if  $\text{IsStable}(nreq(i), Vdisturbed(i), \Theta) = \text{FALSE}$  then
13:        return TRUE
14:      end if
15:    end if
16:  end for
17: end for
18: return FALSE

```

1. Physical Motivation: Input Pin Disturbance and Boundary-Condition Integrity

F6 filter is used to retain the integrity of input boundary conditions. In SiDB standard cell template, the assumption made about the input BDL pairing is that these are externally imposed by input drivers and they should not have their behavior affected by the presence of canvas SiDBs. A layout is said to be unfavorable if the chosen canvas SiDBs exert too much electrostatic back action on the input pins.

In the implementation, the external driver is not modeled explicitly. Instead, the input pins are treated as fixed boundary conditions, and F6 applies a conservative bound on the worst-case canvas-induced disturbance. Let A be the set of selected canvas SiDBs and let $\phi(i, a) > 0$ denote the screened Coulomb potential magnitude between an input-pin site i and a canvas site a . The worst-case signed canvas disturbance is obtained by assuming that all canvas SiDBs are negatively charged:

$$D_i^{wc}(A) = - \sum_{a \in A} \phi(i, a)$$

The default implementation rejects the layout if the magnitude of this worst-case disturbance exceeds a configurable input-disturbance threshold τ_{in} : $|D_i^{wc}(A)| > \tau_{in}$

If no explicit threshold is provided, τ_{in} is chosen as the width of the neutral stability window induced by the physical parameters. Optionally, a stricter mode can be enabled in which the disturbed local potential is checked directly against the population-stability condition for the required input charge state

2. Pruning Condition

A layout $L = S \cup A$ is eliminated if, for at least one input vector $\vec{x}$, the selected canvas SiDBs can exert an excessively large electrostatic back-action on any input-pin SiDB. In this framework, the input BDL pairs are treated as fixed boundary conditions imposed by an external driver. Therefore, instead of explicitly modeling the driver potential, the implemented F6 filter bounds the worst-case disturbance generated by the canvas.

Let A denote the set of selected canvas SiDBs, and let i be an SiDB site belonging to an input pin, We denote by $\phi(i, a) = \frac{k_c}{\epsilon_r d_{ia}} \exp\left(-\frac{d_{ia}}{\lambda_{TF}}\right)$ the positive screened Coulomb potential magnitude between the input site i and a canvas site $a \in A$, where d_{ia} is their Euclidean distance. worst-case canvas-induced disturbance is obtained by assuming that all canvas SiDBs are negatively charged. Under the sign convention used in the implementation, this gives the signed disturbance $D_i^{wc}(A) = - \sum_{a \in A} \phi(i, a)$

The default implementation of F6 discards a layout if the magnitude of this worst-case disturbance exceeds a predefined input-disturbance threshold τ_{in} :

Discard L if $\exists \vec{x} \in \mathbb{B}^n, \exists i \in S_{in}(\vec{x}) : |D_i^{wc}(A)| > \tau_{in}$

Here, $S_{in}(\vec{x})$ denotes the set of input-pin SiDBs under the applied input pattern $\vec{x}$. In the implementation, if τ_{in} is not specified explicitly, it is chosen as the width of the neutral stability window induced by the physical parameters: $\tau_{in} = \left| -\frac{\mu^-}{q_e} + \frac{\mu^+}{q_e} \right|$

Equivalently, this threshold corresponds to the separation between the two charge-transition potential boundaries. This makes F6 a conservative guard against layouts in which canvas SiDBs are placed too close to the input structures and may disturb the externally imposed input encoding.

Optionally, a stricter population-stability mode can be used, In that case, the base potential generated by the skeleton and the imposed I/O assignment is computed first, and the disturbed potential is defined as

$$V_{local,i}^{disturbed} = V_{base,i} + D_i^{wc}(A)$$

The layout is then discarded if the required input charge state n_i^{req} is not population-stable under $V_{local,i}^{disturbed}$:

$$\text{Discard } L \text{ if } \text{IsStable}(n_i^{req}, V_{local,i}^{disturbed}, \Theta) = FALSE$$

However, unless explicitly enabled, the implemented F6 filter uses the bounded-disturbance criterion $|D_i^{wc}(A)| > \tau_{in}$, not the full input population-stability check.

The set of layouts surviving F6 in the default implementation can therefore be written as

$$L_d^{(6)} = \left\{ L \in L_d^{(5)} \mid \forall \vec{x} \in \mathbb{B}^n, \forall i \in S_{in}(\vec{x}) : |D_i^{wc}(A)| \leq \tau_{in} \right\}$$

The computational complexity of the default F6 check is $O(2^n \cdot n_p \cdot d)$, where n_p is the number of SiDB sites in the input-pin structures and d is the number of selected canvas SiDBs. In the optional strict population-stability mode, the cost increases to $O(2^n \cdot n_p \cdot (|S| + d))$, because the base skeleton potential must also be evaluated. Thus, F6 remains a polynomial-cost filter that preserves the integrity of the input boundary conditions before the more expensive enumerative pruning stages are invoked.

3. Completeness, Soundness, and Heuristic Nature of the Filter

Theorem 6 (Conditional Completeness of Input-Disturbance Pruning)

Let τ_{in} be a certified input-disturbance bound such that any valid layout must satisfy

$|D_i^{wc}(A)| \leq \tau_{in}$, for every input-pin SiDB i and every input pattern x . Then F6 is complete

Proof

F6 rejects a layout only if there exist an input-pin site i for which $|D_i^{wc}(A)| > \tau_{in}$

By the certified input-integrity assumption, no valid layout may exceed this disturbance threshold, Therefore the rejected layout cannot belong to $\mathcal{L}_{valid}$. Hence F6 is complete under the certified disturbance-bound assumption.

In current prototype, τ_{in} is a conservative configurable threshold. Therefore F6 is best interpreted as a bounded back-action heuristic unless τ_{in} is independently calibrated.

II B.3-F7: Electrostatic Path Connectivity Pruning

F7_ElectrostaticPathConnectivityPruning

Input: Layout $L = S \cup A$, Skeleton S , physical parameters Θ
Output: TRUE iff no effective electrostatic path exists from the input region to the output region; FALSE otherwise

```

1:  $V \leftarrow$  all SiDB sites in  $L$ 
2:  $reff \leftarrow$  EffectiveElectrostaticConnectivityRadius( $\Theta$ )
3:  $E \leftarrow \emptyset$ 
4: for all pairs  $(v_i, v_j) \in V$  with  $i < j$  do
5:   if EuclideanDistance( $v_i, v_j$ )  $\leq reff$  then
6:      $E \leftarrow E \cup \{(v_i, v_j)\}$ 
7:   end if
8: end for
9:  $G_{conn} \leftarrow (V, E)$ 
10:  $V_{in} \leftarrow$  vertices corresponding to input-pin SiDBs in  $S$ 
11:  $V_{out} \leftarrow$  vertices corresponding to output-pin SiDBs in  $S$ 
12: if  $V_{in} = \emptyset$  or  $V_{out} = \emptyset$  then
13:   return TRUE
14: end if
15:  $V_{reachable} \leftarrow$  BFS( $G_{conn}, V_{in}$ )
16: if  $V_{reachable} \cap V_{out} = \emptyset$  then
17:   return TRUE
18: end if
19: return FALSE

```

1. Physical Principle: Information Propagation via Localized Interactions

In F7, a model of electrostatic coupling is developed using the theory of graphs. The model consists of a set of vertices representing the SiDB sites, and edges are drawn between each pair of vertices within an "effective" electrostatic interaction radius. This model is valid only if the resulting graph contains a path between the input terminal region and the output terminal region. In other words, F7 eliminates layouts that are electrostatically disconnected and hence cannot propagate logical influence from input to output.

The filter F7 encapsulates this physical requirements, which is that the information should be able to propagates between the inputs and output of the device. In an SiDB system, this information propagation is not through a medium, but through a series of localized electrostatic interactions. The influence of an SiDB decays exponentially with distance, governed by the Yukawa potential's $e^{-d/\lambda_{TF}}$ factor. This localized nature of the interaction allows us to define an effective interaction radius, r_{eff} . Beyond this radius, the electrostatic potential contribution from one SiDB to another is considered negligible compared to thermal energy scales or other system noise. A common and physically justified choice for this radius is a multiple of the Thomas-Fermi screening length, such as $r_{\text{eff}} = 3\lambda_{TF}$, where the potential is attenuated by over 95%.

If the SiDBs on the canvas are placed in a disordered or sparse manner, they might not be able to create a "bridge" of electrostatic influence between the input and output pins. F7 aims to detect and eliminate such layouts that are electrostatically disconnected.

F7 should not be interpreted as a proof of information propagation. It merely rejects layouts that are clearly disconnected under a conservative electrostatic reachability model.

2. Graph-Theoretic Model of Electrostatic Conectivity

To analyzes connectivity, we model the entire SiDB layout $L = S \cup A$ as an undirected connectivity graph, $G_{\text{conn}} = (V, E)$, the set of vertices V is the set of all SiDB positions in the layout $V = \{\vec{r}_i \mid i \in L\}$

And An edge exists between two vertices if their corresponding SiDBs are close enough to interact meaningfully. The edge set E is defined as $E = \{(\vec{r}_i, \vec{r}_j) \mid \vec{r}_i, \vec{r}_j \in V, i \neq j, \text{ and } \|\vec{r}_i - \vec{r}_j\|_2 \leq r_{\text{eff}}\}$

The problem of information propagation is now transformed into a standard graph-theoretic connectivity problem. Let $V_{\text{in}} \subset V$ be the set of vertices corresponding to the input pins and $V_{\text{out}} \subset V$ be the set of vertices corresponding to the output pins. For the gate to be functional, there must exist at least one path in G_{conn} from a vertex in V_{in} to a vertex in V_{out}

3. Pruning Algorithm and Condition

Algorithm uses a graph traversal method, such as Breadth-First Search (BFS) or Depth-First Search (DFS), to determine if the input and output regions are in the same connected component of the graph

Algorithm:

1. Construct the connectivity graph $G_{\text{conn}} = (V, E)$ for given layout L

2. Define the set of source vertices as V_{in}
3. Perform a graph traversal (e.g., BFS) starting simultaneously from all vertices in V_{in} , This process will discover all vertices that are reachable from the input region ,Let this set be $V_{reachable}$

$$V_{reachable} = \{v \in V \mid \exists u \in V_{in} \text{ such that there is a path from } u \text{ to } v \text{ in } G_{conn}\}$$

4. Check for a intersection between the set of reachable vertices and set of output vertices

Pruning Condition:

A layout L is discarded if the output region is unreachable from the input region , Discard layout L if $V_{reachable} \cap V_{out} = \emptyset$, The set of layouts that survive F7 is formally defined as:

$L_d^{(7)} = \{L \in L_d^{(6)} \mid \text{Reaches}(L, V_{in}, V_{out})\}$, where $\text{Reaches}(L, U, W)$ is a predicate that is true if there is a path in the connectivity graph of L from a vertex in set U to a vertex in set W

The computational complexity of this filter per layout is dominated by the construction of the adjacency list for the graph, which is $O(|L|^2) = O((|S| + d)^2)$, followed by a graph traversal of complexity $O(|V| + |E|) = O(|L|^2)$. This polynomial-time check is highly efficient compared to enumerative methods. It effectively removes layouts that are structurally incapable of transmitting information, regardless of the specific charge configurations, thereby pruning a significant portion of the search space based on fundamental connectivity requirements

4. Completeness, Soundness, and Heuristic Nature of the Filter

Theorem 7 (Conditional Completeness of Electrostatic Path Connectivity)

Let G_{conn} is the effective electrostatic connectivity graph constructed with radius r_{eff} , Assume the following finite-range propagation condition holds: if there is no path in G_{conn} from the input region V_{in} to the output region V_{out} , then the maximum possible input induced output perturbation is below the minimum signal distinguishability margin ϵ_{out} required to affect the output BDL state

Formally, assume $\neg \text{Reaches}(L, V_{in}, V_{out}) \implies |\Delta V_{out}| < \epsilon_{out}$

and that any valid layout requires $|\Delta V_{out}| \geq \epsilon_{out}$ for at least one input-dependent output transition
Then F7 is complete

Proof

F7 rejects L only if no path exist from V_{in} to V_{out} in G_{conn} , By the finite-range propagation assumption, this imply that the input region cannot induce an output perturbation large enough to affect the output BDL state. Therefore the layout cannot transmit the required logical influence from input to output and cannot realize the target Boolean function. Hence $L \notin \mathcal{L}_{valid}$, The completeness of F7 depend on the finite range propagation assumption and the chosen r_{eff} , Without such a certified margin, F7 should be treated as a structural heuristic rather than an unconditional complete filter

II B.4-F8: Output Potential Bound Pruning

F8: OutputPotentialBoundPruning

Input: Layout $L = S \cup A$, Boolean function f , Skeleton S , physical parameters Θ
Output: TRUE iff the required output charge state is incompatible with attainable output-potential bounds; FALSE otherwise

```

1: for all input pattern  $x \in B^n$  do
2:    $y \leftarrow f(x)$ 
3:   for all output site  $o \in \text{OutputPins}(S)$  do
4:      $q_{\text{req}} \leftarrow \text{RequiredOutputCharge}(o, y)$ 
5:      $V_{\text{base}} \leftarrow \text{FixedSkeletonIOPotential}(o, S, x, y, \Theta)$ 
6:      $V_{\text{canvas\_min}} \leftarrow -\sum_{a \in A} \phi(o, a)$ 
7:      $V_{\text{canvas\_max}} \leftarrow 0$ 
8:      $V_{\text{tot\_min}} \leftarrow V_{\text{base}} + V_{\text{canvas\_min}}$ 
9:      $V_{\text{tot\_max}} \leftarrow V_{\text{base}} + V_{\text{canvas\_max}}$ 
10:    if  $[V_{\text{tot\_min}}, V_{\text{tot\_max}}] \cap W(q_{\text{req}}) = \emptyset$  then
11:      return TRUE
12:    end if
13:  end for
14: end for
15: return FALSE

```

$\phi(o, a)$ is the positive screened Coulomb potential magnitude. Θ denotes the physical parameters.

1. Physical Principle: Bounded Perturbation Theory for Output Stability

Filter F8 is the test applied to ensure that the required output charge state is feasible. The potential level at the output pin SiDB is based on the combination of two factors: a constant factor from the skeleton and a variable factor from the selected canvas SiDBs. In case the entire available range of potential levels does not lie within the population stability window for the required output charge state, the required output charge state cannot be achieved.

For a given input vector $\vec{x}$, the Boolean function determines the required output $y = f(\vec{x})$. For an output pin SiDB site i , the local potential can be decomposed as $V_{\text{local},i} = V_{\text{base},i}(\vec{x}) + V_{A,i}(C_A)$

where $V_{\text{base},i}(\vec{x})$ is the fixed potential generated by the imposed skeleton I/O charge configuration for input $\vec{x}$ and output $f(\vec{x})$, excluding site i itself. The term $V_{A,i}(C_A)$ is the variable contribution from the canvas SiDBs, which depends on the unknown canvas charge configuration $C_A \in \{-1, 0\}^d$. Instead of enumerating all 2^d canvas charge configurations, F8 computes lower and upper bounds on $V_{A,i}$

2. Formulation of Potential Bounds

Let $\phi(i, a)$ denote the positive screened Coulomb potential magnitude between output site i and canvas site $a \in A$: $\phi(i, a) = \frac{k_c}{\epsilon_r d_{ia}} \exp\left(-\frac{d_{ia}}{\lambda_{TF}}\right)$

Under the sign convention used in the implementation, the signed contribution of a canvas SiDB with charge $n_a \in \{-1, 0\}$ is $n_a \phi(i, a)$. Hence, a neutral canvas SiDB contributes 0, whereas a negatively charged canvas SiDB contributes $-\phi(i, a)$. Therefore total canvas contribution is bounded by

$$V_{A,i}^{\min} = -\sum_{a \in A} \phi(i, a), \quad V_{A,i}^{\max} = 0$$

The minimum is attained when all selected canvas SiDBs are negatively charged, and the maximum is attained when all of them are neutral. Thus the attainable interval for the total local potential at output site i is $\mathcal{V}_i(\vec{x}) = [V_{\text{base},i}(\vec{x}) + V_{A,i}^{\min}, V_{\text{base},i}(\vec{x}) + V_{A,i}^{\max}]$

or equivalently, $\mathcal{V}_i(\vec{x}) = [V_{\text{base},i}(\vec{x}) - \sum_{a \in A} \phi(i, a), V_{\text{base},i}(\vec{x})]$

3. Pruning Condition and Set-Theoretic Formulation

Let $T^- = -\frac{\mu^-}{q_e}, T^+ = -\frac{\mu^+}{q_e}$. The population-stability windows used by the implementation are

$$\begin{aligned} W(-1) &= [T^-, +\infty), \\ W(0) &= [\min(T^+, T^-), \max(T^+, T^-)]. \end{aligned}$$

For each input pattern $\vec{x}$, output assignment $y = f(\vec{x})$, and output-pin SiDB site i with required charge $n_i^{\text{req}}(y)$, F8 checks whether the attainable potential interval intersects the required stability window: $\mathcal{V}_i(\vec{x}) \cap W(n_i^{\text{req}}(y)) \neq \emptyset$

The layout is discarded if this intersection is empty:

$$\text{Discard } L \text{ if } \exists \vec{x} \in \mathbb{B}^n, \exists i \in S_{\text{out}}(f(\vec{x})): \mathcal{V}_i(\vec{x}) \cap W(n_i^{\text{req}}) = \emptyset$$

Equivalently, let $v_{\min} = V_{\text{base},i}(\vec{x}) - \sum_{a \in A} \phi(i, a)$, $v_{\max} = V_{\text{base},i}(\vec{x})$

Then:

- If $n_i^{\text{req}} = -1$, the layout is discarded if $v_{\max} < T^-$
- If $n_i^{\text{req}} = 0$, the layout is discarded if $v_{\max} < \min(T^+, T^-)$ or $v_{\min} > \max(T^+, T^-)$

Set of layouts surviving F8 is therefore

$$L_d^{(8)} = \left\{ L \in L_d^{(7)} \mid \forall \vec{x} \in \mathbb{B}^n, \forall i \in S_{\text{out}}(f(\vec{x})): \mathcal{V}_i(\vec{x}) \cap W(n_i^{\text{req}}) \neq \emptyset \right\}$$

implementation has complexity $\mathcal{O}(2^n m_{\text{out}}(|S| + d))$, where m_{out} is the number of output pin SiDB sites. And If the fixed skeleton contribution is precomputed, this reduces to $\mathcal{O}(2^n m_{\text{out}} d)$

4. Completeness, Soundness, and Heuristic Nature of the Filter

Theorem 8 (Completeness of Output-Potential Bound Pruning)

For a layout L and an output site i , let $\mathcal{V}_i(x)$ be the interval containing all attainable local potentials at i over all canvas charge assignments, Let $W(n_i^{\text{req}})$ be the population stability window of the required output charge state, If $\mathcal{V}_i(x) \cap W(n_i^{\text{req}}) = \emptyset$, then no canvas charge assignment can realize the required output charge state at site i

Proof

By construction, F8 computes lower and upper bounds such that for every admissible canvas charge assignment c_A , $V_{\text{local},i}(x, c_A) \in \mathcal{V}_i(x)$, If $\mathcal{V}_i(x)$ has empty intersection with the required stability window $W(n_i^{\text{req}})$, then no possible value of $V_{\text{local},i}$ can lie inside $W(n_i^{\text{req}})$, Therefore the required charges state cannot be population stable at site i for any canvas charge assignments, Hence the layout cannot realize the required output states for input x and is not valid

II B.5-F9: Electrostatic Pressure at Output Pruning

F9 ElectrostaticPressurePruning

```

1: Input: Layout  $L$ , Boolean function  $f$ , Skeleton  $S$ ,
   Physical parameters  $\Theta$ 
2: Output: TRUE iff output BDL pair cannot be biased
   in the required direction; FALSE otherwise
3: for all input pattern  $x \in \mathbb{B}^n$  do
4:   for all output BDL pair  $O_\beta \in \text{OutputBDLPairs}(S)$ 
   do
5:      $\Delta V_{\text{skel}} \leftarrow \text{SkeletonPressureAtOutputPair}(O_\beta, x, f(x))$ 
6:      $\Delta V_{\text{canvas\_min}} \leftarrow \text{MinimumCanvasPressure}(O_\beta, L, P)$ 
7:      $\Delta V_{\text{canvas\_max}} \leftarrow \text{MaximumCanvasPressure}(O_\beta, L, P)$ 
8:      $\Delta V_{\text{min}} \leftarrow \Delta V_{\text{skel}} + \Delta V_{\text{canvas\_min}}$ 
9:      $\Delta V_{\text{max}} \leftarrow \Delta V_{\text{skel}} + \Delta V_{\text{canvas\_max}}$ 
10:    if RequiredOutputPressureSignIsImpossible( $O_\beta, x, f(x), \Delta V_{\text{min}}, \Delta V_{\text{max}}$ )
    then
11:      return TRUE
12:    end if
13:  end for
14: end for
15: end for
16:
17: return FALSE

```

1. Physical Principle: Differential Potential and BDL State Encoding

Filter F9 extend output feasibility analysis from absolute potentials at individual sites to the differential potential across an output BDL pair. Let an output BDL pair consist of two SiDB sites at positions r_a and r_b . We define the electrostatic pressure as $\Delta V = V_{\text{local}}(r_a) - V_{\text{local}}(r_b)$

Under the BDL convention used in implementation, a bit value 0 requires that a negative pressure be attainable, i.e., $\Delta V < 0$, whereas a bit value 1 requires that a positive pressure be attainable, i.e., $\Delta V > 0$. Therefore, a layout can be pruned if the range of achievable output pressure values cannot contain the sign required by the target output bit.

2. Formulation of Electrostatic Pressure Bounds

The total pressure can be decomposed into a fixed skeleton contribution and a variable canvas contribution: $\Delta V = \Delta V_S + \Delta V_A$ Where $\Delta V_S = V_{S'}(a) - V_{S'}(b)$

is computed from the fixed skeleton I/O charge configuration, excluding the output BDL pair under evaluation

Let $\phi(a, j)$ and $\phi(b, j)$ denote the positive screened Coulomb potential magnitudes between a canvas site $j \in A$ and the two output sites a and b , respectively. For a canvas charge $n_j \in \{-1, 0\}$, the pressure contribution of canvas site j is $\Delta V_j(n_j) = n_j[\phi(a, j) - \phi(b, j)]$

For the negative canvas state $n_j = -1$, this becomes $\Delta V_j(-1) = \phi(b, j) - \phi(a, j)$

Since each canvas SiDB can either contribute $\Delta V_j(-1)$ or 0, the minimum and maximum possible canvas pressure contributions are $\Delta V_A^{\min} = \sum_{j \in A} \min(0, \Delta V_j(-1))$, $\Delta V_A^{\max} = \sum_{j \in A} \max(0, \Delta V_j(-1))$

Thus the achievable output pressure interval is $P_{\text{out}}(x) = [\Delta V_S(x) + \Delta V_A^{\min}, \Delta V_S(x) + \Delta V_A^{\max}]$

3. Pruning Condition and Set-Theoretic Formulation

Let the required output bit be b_{req} , and let $\eta \geq 0$ denote the pressure margin used by implementation. In the default convention:

- if $b_{\text{req}} = 0$, a negative pressure must be attainable
- if $b_{\text{req}} = 1$, a positive pressure must be attainable

Therefore:

- For $b_{\text{req}} = 0$, the layout is discarded if the achievable interval cannot reach a sufficiently negative value: Discard if $\min(\mathcal{P}_{\text{out}}(x)) \geq -\eta$
- For $b_{\text{req}} = 1$, the layout is discarded if the achievable interval cannot reach a sufficiently positive value: Discard if $\max(\mathcal{P}_{\text{out}}(x)) \leq \eta$

For the common case $\eta = 0$, these conditions reduce to:

$$b_{\text{req}} = 0: \min(\mathcal{P}_{\text{out}}(x)) \geq 0, \quad b_{\text{req}} = 1: \max(\mathcal{P}_{\text{out}}(x)) \leq 0$$

The set of layouts surviving F9 is

$$L_d^{(9)} = \left\{ L \in L_d^{(8)} \mid \forall x \in \mathbb{B}^n, \forall k: \mathcal{P}_{\text{out},k}(x) \text{ contains the sign required by } f_k(x) \right\}$$

The computational complexity of the implemented F9 check is $O(2^n \cdot m_{\text{out}} \cdot (|S| + d))$,

where m_{out} is the number of output BDL pairs. If the fixed skeleton pressure is precomputed, this reduces to $O(2^n \cdot m_{\text{out}} \cdot d)$

4. Completeness, Soundness, and Heuristic Nature of the Filter

Theorem 9 (Completeness of Output-Pressure Sign Pruning)

Let $\mathcal{P}_{\text{out},k}(x)$ be the interval containing all attainable electrostatic pressure values ΔV across output BDL pair k over all canvas charge assignments. Suppose the required output bit requires a pressure sign $\sigma \in \{\text{negative}, \text{positive}\}$. If $\mathcal{P}_{\text{out},k}(x)$ contains no value with the required sign, then the layout cannot realize the required output bit for input x

Proof

F9 computes bounds such that every possible canvas charge assignment produces an output pressure value inside $\mathcal{P}_{\text{out},k}(x)$. If the interval contains no value with the sign required by the target BDL output bit, then no possible canvas charge state can bias the output pair in the required direction. Therefore the required output bit is unattainable, and the layout cannot be valid

II B.6-F10: Energy Lower Bound Pruning

F10_EnergyLowerBoundPruning

Input: Layout $L = S \cup A$, Boolean function f , Skeleton S , physical parameters Θ

Output: TRUE iff an incorrect I/O state is energetically favored by the neutral-canvas reference test; FALSE otherwise

```

1:  $\gamma \leftarrow \text{EnergyBoundMargin}(\Theta, d)$ 
2:  $C_A^0 \leftarrow$  all-neutral canvas charge assignment
3: for all input pattern  $x \in B^n$  do
4:    $y_{\text{cor}} \leftarrow f(x)$ 
5:    $E_{\text{cor}} \leftarrow E(C_S(x, y_{\text{cor}}) \cup C_A^0)$ 
6:   for all incorrect output assignments  $y_{\text{inv}} \neq y_{\text{cor}}$  do
7:      $E_{\text{inv}} \leftarrow E(C_S(x, y_{\text{inv}}) \cup C_A^0)$ 
8:     if  $E_{\text{cor}} > E_{\text{inv}} + \gamma$  then
9:       return TRUE
10:    end if
11:  end for
12: end for
13: return FALSE

```

In the implementation, $\gamma = \max(\text{energy_tolerance}, \text{energy_bound_margin})$, and if $\text{energy_bound_margin}$ is not specified, $\gamma = |\mu - |d|$.

1. Physical Principle: Analytical Detection of Energetically Suspicious Layouts

Filter F10 is the pruning step based on analysis of energy, which acts as a cheap alternative to the more costly step done using the I/O signal instability test in F12. This filter is aimed at identifying configurations that have an unstable output assignment due to lower energy favoring this configuration.

Rather than enumerating all 2^d canvas charge states, the implementation evaluates the total electrostatic energy using the all neutral canvas assignment $C_A^0 = \{n_a = 0 \mid a \in A\}$

For a given input vector x , let $y_{\text{cor}} = f(x)$ denote the correct output assignment. , The reference energy of the correct state is defined as $E_{\text{cor}}^0(x) = E(C_S(x, y_{\text{cor}}) \cup C_A^0)$

For each incorrect output assignment $y_{\text{inv}} \neq y_{\text{cor}}$, the corresponding reference energy is

$$E_{\text{inv}}^0(x, y_{\text{inv}}) = E(C_S(x, y_{\text{inv}}) \cup C_A^0)$$

If an incorrect output assignment has a sufficiently lower reference energy than the correct one, the layout is considered energetically suspicious and is discarded

2. Energy Model

For a charge configuration $C = \{n_i\}$, the total energy used by the implementation is

$$E(C) = \sum_{i < j} n_i n_j \phi(i, j) + N^-(C) \mu^-, \text{ Where } \phi(i, j) = \frac{k_c}{\epsilon_r d_{ij}} \exp\left(-\frac{d_{ij}}{\lambda_{TF}}\right)$$

is the positive screened Coulomb potential magnitude, and $N^-(C)$ is the number of negatively charged SiDBs in configuration C

3. Pruning Condition

Let $\gamma \geq 0$ be a safety margin used to avoid over pruning, In the implementation,

$\gamma = \max(\epsilon_E, \gamma_{\text{bound}})$, where ϵ_E is the numerical energy tolerance. If no explicit bound margin is specified, the implementation uses $\gamma_{\text{bound}} = |\mu^-| d$

F10 pruning condition is ,Discard Lif $\exists x \in \mathbb{B}^n, \exists y_{\text{inv}} \neq f(x) : E_{\text{cor}}^0(x) > E_{\text{inv}}^0(x, y_{\text{inv}}) + \gamma$

Thus F10 removes only those layouts for which the all neutral reference comparison indicates that an incorrect output state is energetically favored by more than the safety margin

The set of layouts surviving F10 can be written as

$$L_d^{(10)} = \left\{ L \in L_d^{(9)} \mid \forall x \in \mathbb{B}^n, \forall y_{\text{inv}} \neq f(x): E_{\text{cor}}^0(x) < E_{\text{inv}}^0(x, y_{\text{inv}}) + \gamma \right\}$$

The default implementations considers incorrect output assignment for the same input pattern.

Optionally, inverted input assignments can also be considered if the corresponding configuration option is enabled

The computational complexity is $O(2^n \cdot 2^m \cdot N^2)$, where m is the number of output bits and

$N = |S| + d$. For single-output gates, this reduces to $O(2^n \cdot N^2)$

4. Completeness, Soundness, and Heuristic Nature of the Filter

Theorem 10 (Conditional Completeness of Energy-Bound Pruning)

Assume that F10 use a certified lower bound E_{cor}^{LB} on the correct state ground energy and a certified upper bound E_{inv}^{UB} on an incorrect state ground energy , If $E_{\text{cor}}^{LB} > E_{\text{inv}}^{UB}$, then the layout is invalid

Proof

Since E_{cor}^{LB} is a lower bound on the correct-state ground energy, $E_{\text{cor}}^{LB} \leq E_{\text{GS}}(C_S^{\text{cor}})$

Since E_{inv}^{UB} is an upper bound on an incorrect-state ground energy

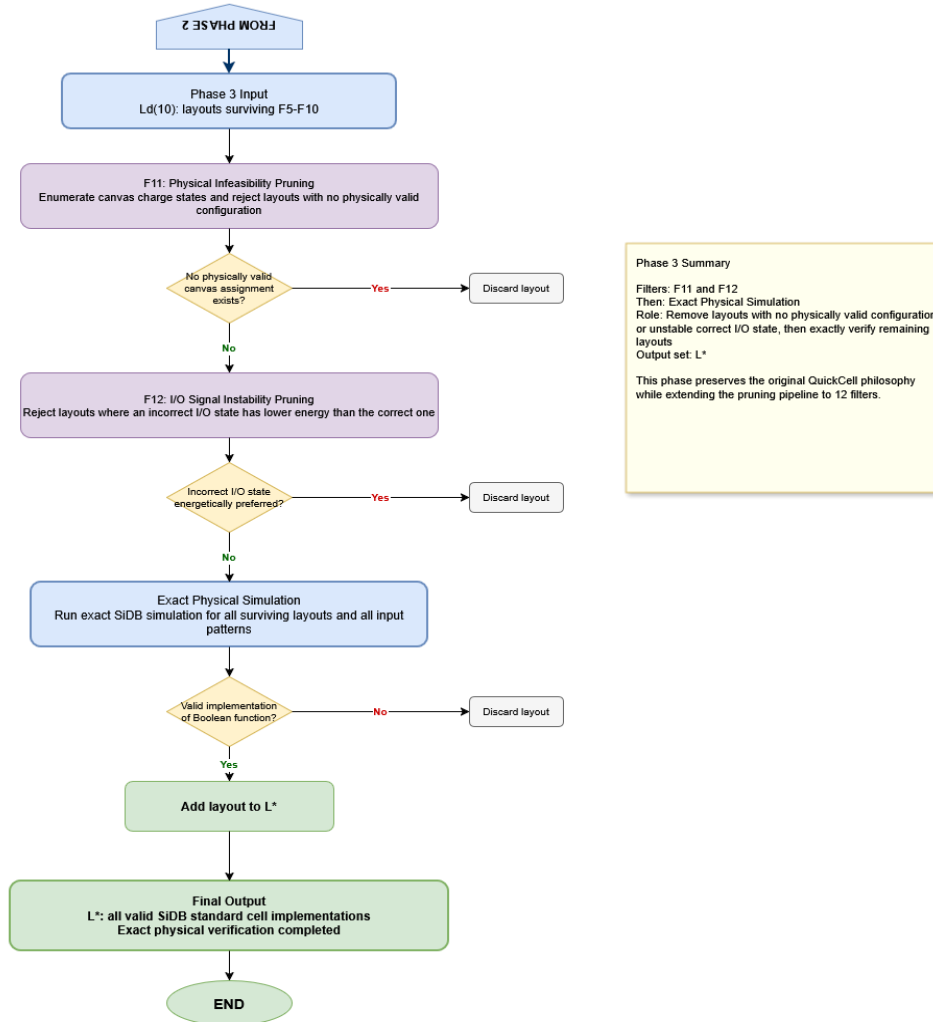
$$E_{GS}(C_S^{\text{inv}}) \leq E_{\text{inv}}^{UB} \quad \text{if } E_{\text{cor}}^{LB} > E_{\text{inv}}^{UB}, \text{ then } E_{GS}(C_S^{\text{cor}}) > E_{GS}(C_S^{\text{inv}})$$

Therefore an incorrect I/O state has lower ground state energy than the correct state, The layout is energetically unstable and cannot be valid

The current implementation uses an all neutral canvas reference energy rather than certified lower and upper bounds. Therefore F10 is not claimed to be unconditionally complete in the prototype, it is treated as an analytical heuristic guard with a safety margin

II C. Phase 3: Enumerative Physical Pruning

Phase 3: Enumerative Physical Pruning (F11-F12) + Exact Physical Simulation
QuickCell upgraded from 3 filters to 12 filters



Phase3 Prune

```
1: Input: Layout  $L$ , Boolean function  $f$ , Skeleton  $S$ ,  
   Canvas  $C$ , Number of canvas SiDBs  $d$ , Physical  
   parameters  $P$   
2: Output: TRUE iff layout  $L$  must be discarded in Phase  
   3; FALSE otherwise  
3: // F11: Physical Infeasibility Pruning  
4: if F11_PhysicalInfeasibilityPruning( $L, f, S, d, P$ ) =  
   TRUE then  
5:  
6:   return TRUE  
7: end if  
8: // F12: I/O Signal Instability Pruning  
9: if F12_IOSignalInstabilityPruning( $L, f, S, d, P$ ) =  
   TRUE then  
10:  
11:   return TRUE  
12: end if  
13:  
14: return FALSE
```

third phase is entered only after the search space has been drastically reduced by the previous two phases. in this phase, number of remaining layouts is small enough so that an actual enumeration over canvas charge states can be done computationally

Phase 3 consists of two pruning filters , F11(Physical Infeasibility Pruning) and F12(I/O Signal Instability Pruning) together, they transform the analytically admissible space into

$$L_d^{(12)} = F_{12} (F_{11}(L_d^{(10)}))$$

II C.1-F11: Physical Infeasibility Pruning

F11_PhysicalInfeasibilityPruning

Input: Layout $L = S \cup A$, Boolean function f , Skeleton S , physical parameters Θ
Output: TRUE iff no physically valid supporting canvas charge assignment exists for at least one input; FALSE otherwise

```
1: for all input pattern  $x \in B^n$  do  
2:    $y \leftarrow f(x)$   
3:    $C\_S^{cor} \leftarrow \text{SkeletonChargeDistributionFor}(S, x, y)$   
4:   feasible  $\leftarrow$  FALSE  
5:   best_energy  $\leftarrow +\infty$   
6:   for all canvas charge assignments  $c\_A \in \{-1,0\}^d$  do  
7:      $C\_L \leftarrow C\_S^{cor} \cup c\_A$   
8:     if IsMetastable( $C\_L, \Theta$ ) = TRUE then  
9:       feasible  $\leftarrow$  TRUE  
10:      best_energy  $\leftarrow \min(\text{best\_energy}, E(C\_L))$   
11:    end if  
12:  end for  
13: if feasible = FALSE then  
14:   if relaxed_enumeration = TRUE then  
15:    continue  
16:   else  
17:    return TRUE  
18:   end if  
19: end if  
20: end for  
21: return FALSE
```

1. Physical Principle: Existential Proof of Metastability

filter F11 is the first explicitly enumerative physical pruning stage. Its purpose is to determine whether, for each input pattern and its required output assignment, there exist at least one canvas charge configuration that can physically support the imposed I/O state

For a given input vector $\mathbf{x}$, the correct output is $\mathbf{y} = \mathbf{f}(\mathbf{x})$. the corresponding skeleton charge configuration is denoted by $\mathbf{C}_S^{cor}(\mathbf{x}, \mathbf{y})$, the canvas SiDBs are not fixed and may each be either neutral or negatively charged. Therefore the set of possible canvas charge assignments is $\mathcal{C}_A = \{-1, 0\}^d$

F11 checks whether there exists at least one assignment $\mathbf{c}_A \in \mathcal{C}_A$ such that the full configuration

$\mathbf{C}_L = \mathbf{C}_S^{cor}(\mathbf{x}, \mathbf{y}) \cup \mathbf{c}_A$ satisfies the physical metastability predicate

in the implementations, the skeleton I/O charge assignment is treated as a fixed boundary condition. By default, the population and configuration stability checks are applied to the free canvas SiDBs, while the fixed skeleton sites impose the required I/O state. A stricter mode can be enabled in which all sites are included in the metastability check

2. Metastability Predicate

The predicate $\text{IsMetastable}(\mathbf{C}_L, \Theta)$ consists of two checks

At first, population stability requires that each checked SiDB site has a charge state compatible with its local electrostatic potential: $n_i \in \text{StableCharges}(V_{local,i}(\mathbf{C}_L), \mu^-, \mu^+)$

second, configuration stability requires that no allowed single electron hop between checked free sites reduces the total electrostatic energy. for a hop from a negatively charged site i to a neutral site j , the configuration is stable only if $\Delta E_{i \rightarrow j} \geq 0$

the implementation evaluates this condition by comparing the total energy before and after possible single electron hops among the checked free sites

3. Pruning Condition

for each input vector $\mathbf{x}$, F11 enumerates all 2^d canvas charge assignments. The layout is discarded if at least one input pattern has no metastable supporting canvas assignment for the correct I/O configuration:

$$\text{Discard } L \text{ if } \exists \mathbf{x} \in \mathbb{B}^n: \{\mathbf{c}_A \in \mathcal{C}_A \mid \text{IsMetastable}(\mathbf{C}_S^{cor}(\mathbf{x}, \mathbf{f}(\mathbf{x})) \cup \mathbf{c}_A, \Theta)\} = \emptyset$$

Equivalently, the layouts surviving F11 are

$$L_d^{(11)} = \left\{ L \in L_d^{(10)} \mid \forall \mathbf{x} \in \mathbb{B}^n, \exists \mathbf{c}_A \in \mathcal{C}_A: \text{IsMetastable}(\mathbf{C}_S^{cor}(\mathbf{x}, \mathbf{f}(\mathbf{x})) \cup \mathbf{c}_A, \Theta) \right\}$$

In case of demonstration mode and when there is no occurrence of metastable configurations, then the implementation will use the minimum energy canvas enumeration state. This is just a demonstration at the level of prototypes and does not have to do anything with physics or physicality.

the number of input patterns is 2^n , and the number of canvas charge assignments is 2^d , For each full configuration, computing local potentials and checking possible single-electron hops costs $O(N^2)$ in the worst case, where $N = |S| + d$

Therefore the strict F11 complexity per layout is $O(2^n \cdot 2^d \cdot N^2)$

this exponential dependence on d makes F11 one of the high cost pruning stages. Its placement after the first ten filters is therefore essential, since only a reduced subset of candidate layouts reaches this enumerative stage.

4.Completeness, Soundness, and Heuristic Nature of the Filter

Theorem 11 (Completeness of Physical Infeasibility Pruning in Strict Mode)

Assume strict physical feasibility requires that for every input x , the correct skeleton I/O assignment $C_S^{\text{cor}}(x, f(x))$ admits at least one metastable canvas charge assignment $c_A \in \{-1, 0\}^d$. If F11 rejects a layout, then the layout is not valid

Proof

F11 rejects L , only if there exists an input x such that no canvas charge assignment c_A satisfies

$\text{IsMetastable}(C_S^{\text{cor}}(x, f(x)) \cup c_A, \Theta)$, But a valid layout must support the correct I/O state by at least one physically admissible metastable configuration for every input. Since this condition fails for x , L cannot be valid

this completeness statement applies to strict mode. In relaxed demonstration mode, the implementation may fall back to a minimum energy enumerated state and therefore F11 should not be interpreted as a strict completeness preserving infeasibility proof.

II C.2-F12: I/O Signal Instability Pruning

F12_IOSignalInstabilityPruning

Input: Layout L , Boolean function f , Skeleton S ,
number of canvas SiDBs d , physical parameters Θ
Output: TRUE iff an incorrect output assignment has
lower energy than the correct one; FALSE otherwise

```

1:  $\epsilon_{IO} \leftarrow \max(\text{energy\_tolerance},$ 
 $\text{io\_instability\_margin})$ 
2: for all input pattern  $x \in B^n$  do
3:    $y_{cor} \leftarrow f(x)$ 
4:    $E_{cor} \leftarrow \text{GroundStateEnergy}(L, S, x, y_{cor}, \Theta)$ 
5:   if  $E_{cor} = +\infty$  then
6:     return TRUE
7:   end if
8:   for all output assignment  $y_{alt} \in B^m$  with  $y_{alt}$ 
 $\neq y_{cor}$  do
9:      $E_{alt} \leftarrow \text{GroundStateEnergy}(L, S, x, y_{alt}, \Theta)$ 
10:    if  $E_{alt} < E_{cor} - \epsilon_{IO}$  then
11:      return TRUE
12:    end if
13:  end for
14: end for
15: return FALSE

```

By default, the input x is kept fixed and only output
assignments are varied. If `instability_check_inputs` is
enabled, alternative input assignments are also checked.

1. Physical Principle: Energetic Correctness of the Output State

F12 filter marks the last step of the enumerative pruning before the actual physical verification stage. Although filter F11 ensures that it is possible to assign inputs/outputs in a way that at least one physically allowed canvas charge pattern exists, filter F12 examines if an erroneous assignment becomes energetically favored over a correct assignment.

for a fixed skeleton I/O charge assignment $\mathcal{C}_S(x, y)$, the implementation defines an enumerated ground state energy

$$E_{GS}(x, y) = \min_{c_A \in \mathcal{C}_A} \{E(\mathcal{C}_S(x, y) \cup c_A) \mid \text{IsMetastable}(\mathcal{C}_S(x, y) \cup c_A, \Theta)\}$$

Where $\mathcal{C}_A = \{-1, 0\}^d$ is the canvas charge state space. If no metastable canvas configuration exists, $E_{GS}(x, y) = +\infty$ in strict mode. In relaxed demonstration mode, the implementation may instead fall back to the lowest-energy enumerated canvas configuration; this mode is used only for prototype-level demonstrations.

2. Pruning Condition

For each input vector $x \in \mathbb{B}^n$, let $y_{cor} = f(x)$ be the correct output assignment. F12 first computes

$E_{cor} = E_{GS}(x, y_{cor})$ If $E_{cor} = +\infty$, the layout is infeasible and is discarded

the filter then compares this energy against competing output assignments $\mathbf{y}_{alt} \neq \mathbf{y}_{cor}$. In the default implementation, the input vector is kept fixed and only the output assignment is varied. Optionally, competing input assignments can also be included if the corresponding configuration option is enabled.

Let $\epsilon_{IO} \geq 0$ denote energetic comparison tolerance used by the implementation. The layout is discarded if an incorrect assignment is lower in energy than the correct assignment by more than this tolerance:

Discard $\mathbf{L}$ if $\exists x \in \mathbb{B}^n, \exists y_{alt} \neq f(x): E_{GS}(x, y_{alt}) < E_{GS}(x, f(x)) - \epsilon_{IO}$

In the implementation, $\epsilon_{IO} = \max(\epsilon_E, \eta_{IO})$, where ϵ_E is the numerical energy tolerance and η_{IO} is the optional I/O instability margin.

Therefore, the set of layouts surviving F12 is

$$L_d^{(12)} = \left\{ L \in L_d^{(11)} \mid \forall x \in \mathbb{B}^n, \forall y_{alt} \neq f(x): E_{GS}(x, f(x)) \leq E_{GS}(x, y_{alt}) + \epsilon_{IO} \right\}$$

3. Algorithmic Complexity

For each input pattern, F12 evaluates the correct output assignment and all competing output assignments. For each such assignment, it enumerates all 2^d canvas charge configurations. Each metastability and energy evaluation costs $\mathcal{O}(N^2)$ in the worst case, where $N = |\mathbf{S}| + d$

Thus the default complexity per layout is $\mathcal{O}(2^n \cdot 2^m \cdot 2^d \cdot N^2)$, where m is the number of output bits. If competing input assignments are also enabled, the additional factor is absorbed into the number of skeleton assignments considered.

output of F12 is the set $L_d^{(12)}$, consisting of layouts that pass all geometric, analytical, physical feasibility, and energetic I/O-stability filters. These layouts are then forwarded to the final exact physical verification stage.

4. Completeness, Soundness, and Heuristic Nature of the Filter

Theorem 12 (Completeness of I/O Signal Instability Pruning in Strict Mode)

Assume zero temperature ground state correctness: for every input x , a valid layout must have the correct I/O assignment as energetically no worse than any competing incorrect output assignment. If F12 rejects a layout because there exists $y_{alt} \neq f(x)$ such that $E_{GS}(x, y_{alt}) < E_{GS}(x, f(x)) - \epsilon_{IO}$, then the layout is not valid

Proof

Only if F12 rejects L when the assignment of the incorrect output state results in a lower energy than the assignment of the correct output state by more than the tolerance ϵ_{IO} does F12 reject L . With the ground state hypothesis, the physical system converges to the minimum energy state. The physical system will thus favor the assignment of the incorrect output state over the correct output state

By default, the implementation compares competing output assignments for the same input pattern. If competing input assignments are also enabled, the theorem applies to the enlarged set of skeleton assignments.

above results show that several filters in QuickCell-12 are complete under explicit modeling assumptions, particularly duplicate elimination, admissible symmetry pruning, output-potential bounding, output-pressure sign pruning, and the strict enumerative filters F11 and F12. Other filters, including F5, F6, F7, and F10, are intentionally formulated as conservative or heuristic pruning guards in the current prototype. For these filters, completeness depend on calibrated thresholds or certified physical bounds. Therefore, the framework distinguishes between formally complete pruning steps and heuristic candidate-reduction steps.

II-D. Final Candidate Verification and External SiQAD Validation

1. Rationale: From Necessary Conditions to Final Candidate Validation

Even when all twelve filters have been successfully applied to a layout, its functional validity should still be validated using a physical simulation engine external to the system. The filters are intended to weed out geometrically, analytically, and physically unlikely layouts prior to the application of the most costly form of validation. Thus, the result of the twelve filters process is initially taken to be,

$L_{\text{cand}} = L_d^{(12)}$, In this work, the candidate layouts surviving F12 were exported and validated using SiQAD. For each exported candidate layout, all input patterns were applied, the corresponding SiQAD simulation was performed, and the output BDL pair was decoded. A candidate was considered externally validated if the decoded output matched the expected Boolean output for all input patterns and no I/O integrity violation was observed, Thus the final externally validated set is defined as

$$L_{\text{SiQAD}} = \{L \in L_d^{(12)} \mid \forall x \in \mathbb{B}^n: \text{SiQAD}(L, x) = f(x)\}$$

Here, $\text{SiQAD}(L, x)$ denotes the logical output decoded from the SiQAD-simulated charge configuration of layout L under input pattern x . Through this process of validation, we can confirm independently that the candidate layouts generated by the QuickCell-12 hierarchy match the desired Boolean function.

final validation stage is computationally expensive because each candidate must be simulated for all input patterns. For a three-input benchmark, this requires $2^3 = 8$ simulations per candidate. However, because QuickCell-12 reduces the candidate set before this stage, the number of required SiQAD simulations is substantially smaller than the number that would be required by a brute force simulation of all candidate layouts

2. Prototype Level Verification Function

Let $L_d^{(12)}$ denote the set of layouts that survives all twelve pruning filters. for each layout $L \in L_d^{(12)}$, the prototype defines an internal verification function $\hat{v}_f(L)$, which evaluates whether the layout remains consistent with the target Boolean function under the internal enumerative physical model.

for each input vector $\vec{x} \in \mathbb{B}^n$, let $y_{\text{cor}} = f(\vec{x})$ be the correct output assignment. the prototype computes the enumerated energy $E_{GS}(\vec{x}, y_{\text{cor}})$ for the correct I/O assignment and compares it against competing incorrect output assignments $y_{\text{alt}} \neq y_{\text{cor}}$, the layout passes the internal verification if

$E_{GS}(\vec{x}, y_{\text{cor}}) \leq E_{GS}(\vec{x}, y_{\text{alt}}) + \epsilon_{IO}$ for all input vectors and all competing output assignments, where ϵ_{IO} is the numerical or user-defined I/O energy margin

thus the prototype-level verified set can be written as $\hat{L}^* = \{L \in L_d^{(12)} \mid \hat{v}_f(L) = 1\}$

This set is useful for comparing pruning effectiveness and candidate reduction, but it should not be confused with the final physically validated set obtained from an independent exact simulator.

3. Optional Exact Physical Verification

for strict physical validation, each candidate layout $L \in L_d^{(12)}$ can be passed to an external exact SiDB simulation engine, such as QuickExact. In that case, the final verification function is defined as

$$v_f(L) = \begin{cases} 1, & \text{if } \forall \vec{x} \in \mathbb{B}^n: \text{Simulate}(L, \vec{x}) = f(\vec{x}), \\ 0, & \text{otherwise.} \end{cases}$$

Here, $\text{Simulate}(L, \vec{x})$ denotes the process of applying the input boundary conditions, computing the physical ground-state charge configuration of the layout using the exact simulator, and reading the logical output from the output BDL pair.

The final set of strictly validated standard-cell implementations is then

$$L^* = \{L \in L_d^{(12)} \mid v_f(L) = 1\}$$

4. Computational Role of the Final Verification Stage

The cost of exact physical verification can be high because each candidate layout must be evaluated for all input patterns. If T_{sim} denotes the cost of one physical simulation for one layout and one input pattern, then the cost of verifying one layout is approximately $2^n \cdot T_{\text{sim}}$

A naive exhaustive approach would require this verification for all layouts in L_d . In contrast, QuickCell-12 invokes the final verification only on the much smaller candidate set $L_d^{(12)}$. Therefore, the reduction factor before final verification is $R_{12F} = \frac{|L_d|}{|L_d^{(12)}|}$. For limited-run or sampling-based experiments, the same quantity should be computed with respect to the number of processed layouts: $R_{12F}^{\text{sample}} = \frac{N_{\text{processed}}}{|L_d^{(12)}|}$

This distinction is important. If the full search space is processed, then $N_{\text{processed}} = |L_d|$. If only a subset of layouts is processed, the reported reduction factor corresponds to the sampled experiment and should be labeled accordingly.

Consequently, the aim of the twelve filters is to minimize the number of layouts that need to undergo exact simulation checks. The better the filtration is done, the fewer simulations will be required.

II-E. Overall Algorithm

this algorithm can thus be considered an improved variant of the original QuickCell method. The principle of pruning prior to expensive verification is maintained, whereas the initial flat three-pruning-filter approach is replaced by a hierarchical twelve-pruning-filter approach involving three stages.

Given a Boolean function f , a fixed skeleton S , a canvas C , and a target number d of canvas SiDBs, the algorithm proceeds as follows:

1. Define the candidate layout space

$$L_d = \{S \cup A \mid A \subseteq P, |A| = d\}$$

2. Apply Phase 1 pruning using F1–F4

3. Apply Phase 2 pruning using F5–F10

4. Apply Phase 3 pruning using F11–F12

5. Return the surviving layouts $L_d^{(12)}$ as the final candidate set L_{cand}

6. Export the surviving candidates to SiQAD compatible layout files

7. For each exported candidate and each input pattern, run SiQAD simulation and decode the output BDL pair

8. Collect the layouts that satisfy the target Boolean function for all input patterns as the externally validated set L_{SiQAD}

Therefore, the original principles of QuickCell are maintained: costly physical verification is postponed until after the aggressive pruning process. Nevertheless, the pruning front end generalizes from three filters to twelve filters, separated into three stages of progressively more physical specific and computationally demanding filters.

Remark on Completeness and Heuristic Filters

however, not all filters within the hierarchy for QuickCell-12 are said to be complete from a strict mathematical standpoint. Filters such as F1 and F2 discard only those layouts that are invalid geometrically or identical to a layout that survived the filtration process. Interval filters like F8 and sign-based pruning of output pressure in filter F9 form part of necessary condition checks supported by analysis.

in contrast, the F5, F6, F7, and F10 filters are based on conservative or heuristic pruning tests. These tests aim at filtering out layouts which are highly unlikely to have any value prior to running more complex enumerative tests. Thus, these filters do not claim to offer any formal completeness proof for

any possible combination of SiDB instances. On the contrary, their performance can only be tested using ablation experiments. Such an observation is critical since the proposed framework should be considered as a practical front-end solution rather than a formal one.

III. Evaluation:

In this section, we provide an evaluation of the suggested QuickCell-12 pruning algorithm. The purpose of such evaluation is to determine the effectiveness of the suggested pruning scheme based on twelve filters against the basic QuickCell approach, which uses only three filters.

the evaluation attempts to answer the following:

1. How well do the twelve filters prune the layout candidates?
2. How much more reduction was possible compared to the initial three-filter QuickCell flow?
3. Which filters make the most contribution to the pruning process?
4. How does each of the three phases of the pruning process behave in terms of the retention ratio, pruning ratio, and reduction factor?
5. How much can the number of remaining layouts at the expensive verification phase be reduced?

the evaluation is organized as follows. Section III-A describes the experimental setup, section III-B presents a filter level ablation analysis of the twelve filters, Section III-C compares the proposed QuickCell-12 flow with the original three filter QuickCell-style flow , section III-D reports phase wise runtime and scalability behavior and Finally, Section III-E discusses the implications of the obtained pruning reductions

III -A. Setup

In order to measure the effectiveness of the proposed QuickCell-12 framework, an evaluation environment was implemented in Python. this environment follows the standard cell design paradigm used in QuickCell: for a Boolean function $f: \mathbb{B}^n \rightarrow \mathbb{B}^m$, a fixed skeleton S with predefined I/O pins and a canvas C of admissible positions are given, Candidate layouts are defined as

$L_d = \{S \cup A \mid A \subseteq P, \mid A \mid = d\}$, where P is the set of admissible canvas positions and d is the number of canvas SiDBs to be placed

for the paper size benchmark configuration used in this evaluation, the canvas contains

$\mid C \mid = 143$ admissible positions, and $d = 4$, canvas SiDBs are selected. Thus, the full combinatorial search space is $\mid L_d \mid = \binom{143}{4} = 16,701,685$

in view of the computational difficulty involved in running a full experiment on the whole search space for all benchmarks, the experiments run were conducted through sampling methods. This means that for each benchmark function, 100,000 layouts were evaluated. The full search space size $\mid L_d \mid$ is still

reported to indicate the size of the original combinatorial design problem, while the filter level and phase level results are computed over the processed subset

the following benchmark functions is considered:

AND3, XOR-AND, OR-AND, ONEHOT, MAJ, GAMBLE, DOT, ITE, AND-XOR, XOR3

Each processed layout are evaluated using the proposed twelve filter hierarchy:

- Phase 1: $F1 - F4$
- Phase 2: $F5 - F10$
- Phase 3: $F11 - F12$

the original QuickCell style baseline is evaluated using the corresponding three pruning filters:

1. Positive Charge Pruning
2. Physical Infeasibility Pruning
3. I/O Signal Instability Pruning

the two flows are measured using the same processed subset of layouts for each benchmark, ensuring that the comparison reveals the influence of the pruning approach and not variations in the pool of candidates.

the physical parameters used in the evaluation are fixed to representative values commonly used in SiDB studies: $\mu^- = -0.31$ eV, $\mu^+ \approx -0.80$ eV, $\lambda_{TF} = 5.0$ nm, $\epsilon_r = 5.6$

The implementation employs the screened Coulomb interaction method using Thomas Fermi screening. The enumeration filters utilize the internal metastability and energy models developed in the prototype. After going through the twelve-stage pruning process, the candidate layouts were exported and subjected to validation in SiQAD. Each exported layout is simulated with respect to all the input patterns, and the corresponding output BDL pair is extracted. A layout is said to be successfully validated if there is a match between the extracted output and the target Boolean function for all input patterns, and there are no I/O pin integrity issues.

For each benchmark instance, the following quantities are recorded:

- full search space size $|L_d|$
- number of processed layouts $N_{\text{processed}}$
- number of layout remaining after each filters $F1, \dots, F12$
- number of layouts remaining after each phases
- phase retention ratio

- phase pruning ratio
- phase reduction factor
- runtime of the limited run experiment
- number of layouts reaching the final candidate stage
- corresponding comparison with the original three filter QuickCell style flow

retention ratio after a filter F_k is defined as $\rho_k = \frac{N_k}{N_{k-1}}$, where N_k is the number of layouts remaining after filter F_k and the pruning ratio is $\pi_k = 1 - \rho_k$

For a phase P_j , the phase retention ratio is $\rho(P_j) = \frac{N_{out}(P_j)}{N_{in}(P_j)}$, and the phase reduction factor is $R(P_j) = \frac{N_{in}(P_j)}{N_{out}(P_j)}$

For the limited run evaluation, the cumulative reduction factor of QuickCell 12 is computed as

$R_{12F}^{sample} = \frac{N_{processed}}{N_{12}}$, where N_{12} is number of layouts surviving all twelve filters. This value quantifies how much the twelve filter hierarchy reduces the number of layouts reaching the final candidate stage within the processed sample.

to compare against the original QuickCell style flow, let N_{QC} denote the number of layouts surviving the three filter baseline and N_{12F} denote the number of layouts surviving the proposed twelve filter pipeline.

the relative candidate reduction is measured as $R_{vsQC} = \frac{N_{QC}}{N_{12F}}$

Similarly, if t_{QC} and t_{12F} denote the measured runtimes of the two flows on the same processed subset, the runtime speedup is $S_{time} = \frac{t_{QC}}{t_{12F}}$

the implementation allows layout level parallelism as well. Since filters have to be applied in sequence, parallelism is achieved within each stage of filter processing through concurrent evaluation of different layouts. The semantics of the filtering hierarchy remain preserved while the computational cost of expensive filters is reduced.

The approach facilitates analysis of the proposed framework from four distinct points of view:

1. filter-level efficacy using the ablation results obtained after each filter
2. phase-level efficacy using the combined reductions provided by Phases 1, 2, and 3
3. performance in comparison with the baseline three-filter QuickCell flow; and
4. runtime and scalability properties during layout-level parallel processing

III -B. Ablation Study of the Twelve Filters

Table I reports the filter level ablation results of the proposed twelve filter pruning framework , for each benchmark, the full combinatorial search space size is $|L_d| = \binom{143}{4} = 16,701,685$

However, the Python prototype evaluation is performed as a limited run experiment over a processed subset of candidate layouts. In the reported experiments, each benchmark is evaluated over

$N_{\text{processed}} = 100,000$ candidate layouts. therefore, the values reported after $F1, \dots, F12$ represent the number of surviving layouts within the processed subset, not a full exhaustive traversal of all 16,701,685 layouts

Starting from $N_{\text{processed}}$, Table I record the number of layouts remaining after each filter $F1$ through $F12$. the final count after $F12$ is denoted as the number of candidate layouts, $|L_{\text{cand}}| = |L_d^{(12)}|$ rather than the final physically validated implementation count $|L^*|$

For each benchmark g , the sample level reduction factor of QuickCell-12 is defined as

$R_{12F}^{\text{sample}}(g) = \frac{N_{\text{processed}}}{N_{12F}(g)}$, where $N_{12F}(g)$ is the number of layouts surviving all twelve filters for benchmark g . The corresponding pruning ratio is $\pi_{12F}^{\text{sample}}(g) = 1 - \frac{N_{12F}(g)}{N_{\text{processed}}}$

Using the values from Table III, the benchmark level reductions are:

Table I : benchmark level reductions

For each benchmark g , the sample-level reduction factor of QuickCell-12 is computed as

$$R_{12F}^{sample}(g) = \frac{N_{processed}}{N_{12F}(g)}, \text{ Using } N_{processed} = 100,000, \text{ we obtain:}$$

Benchmark	$N_{processed}$	N_{12F}	R_{12F}^{sample}	Pruning Ratio
AND3	100,000	2	50000×	99.99%
AND-XOR	100,000	3	33333.33×	99.997%
DOT	100,000	1	100000×	99.9614%
GAMBLE	100,000	2	50000×	99.998%
ITE	100,000	1	100000×	99.999%
MAJ	100,000	3	33333.33×	99.997%
ONEHOT	100,000	2	50000×	99.998%
OR-AND	100,000	3	33333.33×	99.997%
XOR3	100,000	2	50000×	99.998%
XOR-AND	100,000	3	33333.33×	99.997%

These results show that, within the processed subset, QuickCell-12 reduce candidate set by approximately $33333.33 \times$ to $100000 \times$, corresponding to pruning ratios between 99.9614% and 99.999%. the strongest reductions occur in benchmarks where output potential and output pressure filters eliminate a large portion of the candidate set before the enumerative filters are reached

in summary, the results of the ablation study show that the twelve filters are not redundant; rather, they operate on varying levels of specificity to filter out candidate layouts prior to the costly process of validation

III -C. Runtime and Scalability Analysis

table IV summarizes the phase wise behavior of the proposed QuickCell-12 framework. For each phase, the table reports:

- number of input layouts
- number of output layouts
- retention ratio
- pruning ratio
- reduction factor
- and the measured runtime of the limited-run experiment

for a phase P_j , the retention ratio is defined as $\rho(P_j) = \frac{N_{\text{out}}(P_j)}{N_{\text{in}}(P_j)}$

pruning ratio is $\pi(P_j) = 1 - \rho(P_j)$, and the phase reduction factor is $R(P_j) = R(P_j) = \frac{N_{\text{in}}(P_j)}{N_{\text{out}}(P_j)}$

for the limited run evaluation, the total reduction factor is computed with respect to the processed subset: $R_{\text{total}}^{\text{sample}} = \frac{N_{\text{processed}}}{N_{12}}$

The outcomes at various phase indicate the actual computation purpose served by the hierarchy in three stages. Phase I makes general cuts based on structure. The second Phase uses more analysis based constraints to achieve greater reduction of candidates. The third Phase is very physical in its checks to reduce the number of candidates further.

the behavior of the algorithm during run-time also shows that it is a sequential process due to the fact that each of the filters acts on the survivors of the previous filters. However, the architecture allows for concurrency in the application of the filters since the layouts for each filter are independent of each other and hence can be applied in parallel. This allows filters like F11 and F12 to take advantage of multiprocessing.

because the reported evaluation is limited to 100,000 processed layouts per benchmark, the measured runtimes should be interpreted as prototype runtime for the processed subset rather than full exhaustive runtimes for all 16,701,685 possible layouts. Nevertheless the observed reduction factors indicate how the proposed twelve filter hierarchy reduces the verification workload relative to the unpruned candidate set.

Collectively, the phase-wise outcomes validate the underlying philosophy behind QuickCell-12 in that cheap structural filters need to exclude as many potential candidates as possible in order for the more costly enumeration process to follow

III -D. Discussion of Results and Design Implications

the third and final step in the evaluation interprets the observed reduction in search space and the behavior of the run-time in terms of real-world SiDB standard cell synthesis. The primary goal of the QuickCell-12 method being proposed is to minimize the number of layout configurations that must be processed through costly enumeration and verification steps. As such, not only is the number of survivors at the end important, but how quickly invalid layouts are pruned out

If the proposed framework is effective, the expected observations are as follows:

1. **Phase 1** should filter out geometrically or structurally unnecessary candidates at minimal computational cost. For example, it is anticipated that pruning based on symmetry and positive charge will decrease the number of candidate structures prior to applying physics reasoning
2. **Phase 2** will decrease the number of designs that pass through the enumerative filters by utilizing necessary conditions based on physics analysis such as charge-count feasibility, input

disturbance limits, connectivity, potential attainability, output pressure direction, and reference energy conformity

3. **Phase 3** should act as the most physically specific pruning stage. Although it is computationally more expensive than the previous phases, it is applied only to the reduced candidate set produced by Phases 1 and 2.
4. number of layouts reaching the final candidate-verification stage should be substantially smaller than in the original three-filter QuickCell-style flow
5. overall runtime should be reduced because the expensive enumerative checks are applied to fewer layouts

in the current Python prototype, the evaluation is performed as a limited run, sampling based experiment. The comparison with the original QuickCell framework is reported in terms of candidate reduction before external simulation. Nevertheless, to ensure further confidence about the generated solutions, the QuickCell-12 candidates after F12 were exported for validation in SiQAD. In this regard, there is an attempt to distinguish between two distinct evaluation criteria: the pruning comparison evaluates the number of discarded candidates prior to simulation, whereas SiQAD validation assesses the realizability of the generated candidates under simulation

Overall, the expected design implication is that the proposed twelve-filter framework preserves the main philosophy of QuickCell—avoiding expensive verification on obviously invalid layouts—while strengthening the pruning front-end through a more expressive hierarchy of geometric, analytical, and enumerative filters

TABLE III — Limited run ablation results for the proposed QuickCell-12 framework

The column $|L_d|$ reports the full combinatorial search space size, whereas the column Processed reports the number of candidate layouts actually evaluated in the Python prototype. All filter counts after F1–F12 are computed over the processed subset of 100,000 layouts per benchmark. The final column Candidates denotes layouts surviving F12 and should not be interpreted as the strictly validated set L^*

Benchmark	$ L_d $	Processed	After F1	After F2	After F3	After F4	After F5	After F6	After F7	After F8	After F9	After F10	After F11	After F12	Total Time [s]
AND3	16701685	100000	52604	52600	30316	22581	19632	6722	6721	6718	5818	2621	2620	2	1455.4024
XOR-AND	16701685	100000	52616	52600	30318	22584	19637	6724	6721	6718	5718	2590	2580	3	1483.8617
OR-AND	16701685	100000	52607	52605	30310	22587	19636	6723	6721	6718	5718	2590	2590	3	1391.6676
ONEHOT	16701685	100000	52615	52608	30326	22582	19638	6728	6725	6717	5732	2589	2589	2	1268.1678
MAJ	16701685	100000	52608	52600	30317	22581	19639	6729	6723	6717	5717	2588	2581	3	1296.5222
GAMBLE	16701685	100000	52625	52605	30318	22587	19637	6725	6722	6718	5718	2591	2590	2	1076.6238
DOT	16701685	100000	52604	52602	30310	22589	19634	6729	6728	6718	5717	2587	2587	1	1173.0319
AND-XOR	16701685	100000	52615	52609	30316	22588	19633	6729	6729	6717	5716	2589	2580	3	1035.4798
ITE	16701685	100000	52635	52604	30319	22581	19632	6722	6722	6718	5717	2587	2580	1	1495.8019
XOR3	16701685	100000	52615	52608	30313	22580	19631	6728	6723	6717	5732	2589	2580	2	1044.8154

TABLE IV : Phase wise reduction summary for the proposed QuickCell-12 framework

The table gives information on the layouts of the inputs, outputs, the retention ratios, pruning ratios, and the reduction factor for each benchmark in each phase. The "Runtime" column indicates the runtime of the full twelve-filter limited run experiment

benchmark	Phase	Filters	Input Layouts	Output Layouts	Retention Ratio [%]	Pruning Ratio [%]	Reduction Factor	Runtime [s]
AND3	Phase 1	F1-F4	100000	22580	22.58	77.42	4.4287	
	Phase 2	F5-F10	22580	2621	11.6076	88.3924	8.615	
	Phase 3	F11-F12	2621	2	0.0763	99.9237	1310.5	
	Total	F1-F12	100000	2	0.002	99.998	50000	1455.4024
AND-XOR	Phase 1	F1-F4	100000	22580	22.58	77.42	4.4287	
	Phase 2	F5-F10	22580	2589	11.4659	88.5341	8.7215	
	Phase 3	F11-F12	2589	3	0.1159	99.8841	863	
	Total	F1-F12	100000	3	0.003	99.997	33333.3333	1035.4798
DOT	Phase 1	F1-F4	100000	22580	22.58	77.42	4.4287	
	Phase 2	F5-F10	22580	2587	11.457	88.543	8.7283	
	Phase 3	F11-F12	2587	1	0.0386	99.9614	2587	
	Total	F1-F12	100000	1	0.001	99.999	100,000.00	1173.0319
GAMBLE	Phase 1	F1-F4	100000	22580	22.58	77.42	4.4287	
	Phase 2	F5-F10	22580	2591	11.4748	88.5252	8.7148	
	Phase 3	F11-F12	2591	2	0.0772	99.9228	1295.5	
	Total	F1-F12	100000	2	0.002	99.998	50000	1076.6238
ITE	Phase 1	F1-F4	100000	22580	22.58	77.42	4.4287	
	Phase 2	F5-F10	22580	2587	11.457	88.543	8.7283	
	Phase 3	F11-F12	2587	1	0.03865	99.96135	2,587.00	
	Total	F1-F12	100000	1	0.001	99.999	100,000.00	1495.8019
MAJ	Phase 1	F1-F4	100000	22580	22.58	77.42	4.4287	
	Phase 2	F5-F10	22580	2588	11.4615	88.5385	8.7249	
	Phase 3	F11-F12	2588	3	0.1159	99.8841	862.6667	
	Total	F1-F12	100000	3	0.003	99.997	33333.3333	1296.5222
ONEHOT	Phase 1	F1-F4	100000	22580	22.58	77.42	4.4287	
	Phase 2	F5-F10	22580	2589	11.4659	88.5341	8.7215	
	Phase 3	F11-F12	2589	2	0.0772	99.9228	1,294.5	
	Total	F1-F12	100000	2	0.002	99.998	50,000	1268.1678
OR-AND	Phase 1	F1-F4	100000	22580	22.58	77.42	4.4287	
	Phase 2	F5-F10	22580	2590	11.4703	88.5297	8.7181	
	Phase 3	F11-F12	2590	3	0.1158	99.8842	863.3333	
	Total	F1-F12	100000	3	0.003	99.997	33333.3333	1391.6676
XOR3	Phase 1	F1-F4	100000	22580	22.58	77.42	4.4287	
	Phase 2	F5-F10	22580	2589	11.4659	88.5341	8.7215	
	Phase 3	F11-F12	2589	2	0.0772	99.9228	1,294.50	
	Total	F1-F12	100000	2	0.002	99.998	50,000	1044.8154
XOR-AND	Phase 1	F1-F4	100000	22580	22.58	77.42	4.4287	
	Phase 2	F5-F10	22580	2590	11.4703	88.5297	8.7181	
	Phase 3	F11-F12	2590	3	0.1158	99.8842	863.3333	
	Total	F1-F12	100000	3	0.003	99.997	33333.3333	1483.8617

TABLE V: Comparison between the original three-filter QuickCell style flow and the proposed QuickCell-12 pipeline on the same processed subset

Candidates reduction is computed as $NQC/N12F$, where NQC is the number of layouts surviving the three filter baseline and $N12F$ is the number of layouts surviving the twelve-filter pipeline

In this table, given that the evaluation was previously performed for 100,000 samples, but in the paper(QuickCell: Fast Automatic Design of Standard Cells for Silicon Dangling Bond Logic), the evaluation was performed on 16701685 samples, so in order for both results to be comparable, the results obtained in this paper for 100,000 samples were multiplied by 167

Benchmark	METRIC											
	Number of pruning filters			Number of pruning phases			Candidate implementations			Search-space reduction		
	QC-3	QC-12	Improvement	QC-3	QC-12	Improvement	QC-3	QC-12	Improvement	QC-3	QC-12	Improvement
AND3	3	12	4.0x more	1	3	hierarchical	3065	334				
AND-XOR	3	12	4.0x more	1	3	hierarchical	2553	501				
DOT	3	12	4.0x more	1	3	hierarchical	288	167				
GAMBLE	3	12	4.0x more	1	3	hierarchical	4981	334				
ITE	3	12	4.0x more	1	3	hierarchical	356	167				
MAJ	3	12	4.0x more	1	3	hierarchical	3060	501				
ONEHOT	3	12	4.0x more	1	3	hierarchical	1920	334				
OR-AND	3	12	4.0x more	1	3	hierarchical	1085	501				
XOR3	3	12	4.0x more	1	3	hierarchical	1490	334				
XOR-AND	3	12	4.0x more	1	3	hierarchical	3149	501				

Estimated Brute-Force Simulation Runtime

Assuming an exact physical simulation time of $t_{sim} = 0.01$ s

per layout and input pattern, and since all considered functions are three-input functions, $2^n = 2^3 = 8$

For the processed subset, $T_{SOTA}^{sample} = 100,000 \times 8 \times 0.01 = 8000$ s

This corresponds to approximately 2.22hours.

For the full search space,

$$T_{SOTA}^{full} = 16,701,685 \times 8 \times 0.01 = 1,336,134.8 \text{ s}$$

1,336,134.8 s ÷ 86,400 ≈ 15.46 days

Estimated Speedup over Naive Simulation

The estimated speedup of QuickCell-12 over naive simulation on the processed subset is

$$S_{\text{SOTA}}^{\text{sample}}(g) = \frac{T_{\text{SOTA}}^{\text{sample}}}{T_{12F}(g)}$$

Using $T_{\text{SOTA}}^{\text{sample}} = 8000\text{s}$, we obtain:

Table IX

Benchmark	$T_{12F}[\text{s}]$	$S_{\text{SOTA}}^{\text{sample}}$
AND3	1455.4024	5.49×
AND-XOR	1035.4798	7.72×
DOT	1173.0319	6.819×
GAMBLE	1076.6238	7.430×
ITE	1495.8019	5.34×
MAJ	1296.5222	6.17×
ONEHOT	1268.1678	6.306×
OR-AND	1391.6676	5.748×
XOR3	1044.8154	7.656×
XOR-AND	1483.8617	5.391×

This indicates that, under the assumed simulation cost, the QuickCell-12 prototype is estimated to be between 5.34 × and 7.656 × faster than directly simulating the processed subset

III-G. External SiQAD Validation

to validate the candidates produced by the QuickCell-12 pruning pipeline, the layouts surviving $F12$ were exported to SiQAD. For each benchmark, the exported candidate layouts were simulated under all input patterns. For each simulation, the output BDL pair was decoded according to the same encoding convention used in the pruning framework. A candidate was considered SiQAD-valid only if the decoded output matched the expected Boolean output for all input patterns and the I/O pins remained integrity-preserving. The validation results confirmed that the exported post- $F12$ candidates were consistent with the expected Boolean behavior in SiQAD for all evaluated benchmark functions. This external validation supports the correctness of the pruning pipeline by showing that the candidates produced by QuickCell-12 are not merely internally consistent under the prototype model, but also agree with an independent SiQAD simulation workflow

The SiQAD validation step is not applied to the entire raw search space. Instead, it is applied only after pruning, which is precisely the intended role of QuickCell-12: to reduce the number of expensive physical simulations required for standard-cell synthesis.

TABLE 4 — External SiQAD validation of post-F12 QuickCell-12 candidates. Each exported candidate was simulated under all input patterns. A candidate is counted as valid only if all decoded outputs match the target Boolean function and no I/O pin-integrity violation is observed.

Table X

Benchmark	Exported Candidates	Input Patterns	SiQAD Simulations	SiQAD-Valid Candidates	Pass Rate
AND3	20	8	160	20	100%
XOR-AND	20	8	160	20	100%
OR-AND	20	8	160	20	100%
ONEHOT	20	8	160	20	100%
MAJ	20	8	160	20	100%
GAMBLE	20	8	160	20	100%
DOT	20	8	160	20	100%
ITE	20	8	160	20	100%
AND-XOR	20	8	160	20	100%
XOR3	20	8	160	20	100%

IV. Conclusion

In recent years, Silicon Dangling Bond (SiDB) logic has emerged as a promising post-CMOS computing paradigm, motivating the development of efficient design automation methods for atomically precise standard-cell synthesis. The original QuickCell framework demonstrated that pruning is an effective strategy for reducing the cost of automatic SiDB layout search. However, its pruning stage is based on only three filters, which allows a large number of structurally or physically implausible layouts to reach the more expensive enumerative and verification stages.

To address this limitation, this work proposed QuickCell-12, a hierarchical three-phase pruning framework that extends the QuickCell concept from three filters to twelve filters. The proposed framework organizes the pruning process into:

- **Phase 1: Geometric and Structural Pruning,**
- **Phase 2: Analytical Physics-Based Pruning,** and
- **Phase 3: Enumerative Physical Pruning.**

Unlike the original flat three-filter approach, QuickCell-12 introduces a richer hierarchy of necessary conditions, including minimum-distance validity, symmetry reduction, wire-interference avoidance, positive-charge prevention, charge-count feasibility, input-pin disturbance bounds, electrostatic path connectivity, output-potential attainability, output-pressure direction, reference-energy consistency, physical feasibility, and I/O signal stability.

The evaluation of the Python prototype shows that the twelve-filter hierarchy can substantially reduce the number of layouts reaching the final candidate stage. In the limited-run evaluation, each benchmark is evaluated over 100,000 processed candidate layouts, while the full combinatorial search-space size is

$$|L_d| = \binom{143}{4} = 16,701,685.$$

Across the considered 3-input benchmark functions, the proposed QuickCell-12 pipeline reduces the processed candidate set to **between 167 and 501** layouts after all twelve filters. This corresponds to pruning ratios between 99.9614% and 99.999%. These results confirm that the added filters are complementary: early filters remove geometric and structural redundancy, middle-stage filters apply analytical electrostatic constraints, and later filters perform more selective enumerative checks on a reduced candidate set.

The estimated comparison with naive brute-force simulation further demonstrates the importance of pruning. Assuming a simulation cost of 0.01s per layout and input pattern, directly simulating the 100,000-layout processed subset would require approximately 8,000s. In contrast, the QuickCell-12 prototype completes the processed benchmarks in substantially less time, yielding estimated speedups over naive simulation **between 5.34 × and 7.656 ×**, depending on the benchmark.

The main contribution of this work is therefore not a replacement for exact SiDB simulation, but a stronger pruning front-end that reduces the workload passed to expensive verification stages. In the current prototype, the pruning pipeline first produces candidate layouts after F12; these candidates were then externally checked using SiQAD for the evaluated validation set.

. Nevertheless, the results indicate that replacing the original three-filter pruning stage with a structured twelve-filter hierarchy can substantially improve candidate reduction and provides a promising direction for scalable SiDB standard-cell design automation.

In addition to the internal pruning evaluation, the post-F12 candidate layouts were exported and validated using SiQAD. For each evaluated candidate, all input patterns were simulated and the output BDL pair was decoded. The SiQAD validation confirmed that the exported candidates were consistent with the expected Boolean behavior for the evaluated benchmark functions. This external validation strengthens the evidence that the QuickCell-12 pipeline produces physically meaningful candidate layouts rather than only internally consistent pruning results.

V. Future Work

While the presented QuickCell-12 scheme represents an improvement to the pruning performance of the QuickCell-inspired technique, some interesting areas for future research remain.

The first one is the experimental study of the false-negative effect through the independent exact simulator. Several filters used in the suggested filtering scheme are justified either through proven geometrical or interval relationships or heuristic/conditionally complete filters. In the current prototype

implementation of the framework, it is possible to evaluate the false-negative effect relative to its internal model only. The further study should test if the proposed filtering removes any physically correct layout due to its heuristic nature.

Another approach will be the utilization of exact geometric benchmarks based on the template standard cells in SiDB. While current benchmarks utilize paper-size benchmarks to reflect the desired search space size, reproducing the exact results presented by QuickCell will require the inclusion of exact skeleton and canvas coordinates in previous benchmarks. The exact incorporation of these benchmarks will result in a more direct comparison with the earlier research and a more accurate assessment of the proposed filters.

The third avenue would be to optimize for better runtime performance. As the costly filters, particularly the enumerative filters F11 and F12, repeatedly calculate the electrostatic energy and metastability criteria, the next generation of software can take advantage of precomputed potential matrices, vectorized calculations, memoization, just-in-time compilation, GPU computations, or a lower level implementation in C++. This will facilitate extensive calculation on the entire set of 16.7 million layouts.

fourth direction is the refinement of the analytical filters themselves. The proposed twelve filters cover geometric, structural, electrostatic, energetic, and enumerative feasibility aspects, but additional necessary conditions may further improve pruning. Possible extensions include more accurate local electric-field bounds, defect-aware perturbation models, temperature-dependent stability margins, improved wire-interference models, and tighter many-body energy bounds

fifths direction is robustness driven post-processing. In the present approach, attention has been centered on minimizing the search space of potential candidates along with finding energetically feasible configurations. But, the actual design process for SiDB standard cells needs considerations regarding robustness towards manufacturing variations, parameter uncertainties, temperature effects, operational domain limitations, and even more integration-related concerns

In addition, QuickCell-12 itself can be extended to become a pruning-and-optimization flow. After the twelve filter process has thinned out the combinatorial search space down to the promising subset of candidates, various optimization techniques, such as Genetic Algorithms, may be used to further refine this set with an eye toward optimizing other criteria such as:

- minimum bounding box area
- minimum number of SiDBs on the canvas
- maximum energy margin between correct/incorrect states
- robustness to changes in physical parameters
- fault tolerance, and
- size of operational domain

Thus, we achieve the best of both worlds by maintaining the power of deterministic filtering and allowing for multi-criteria optimization of the SiDB standard-cell candidates